Filter in Frequency Domain

Dan Wu

danwu.bme@zju.edu.cn

Previously

- 2D Fourier Transform
  - *spatial domain* to *frequency domain*

- **2D Sampling and Aliasing**
  - 2-D *Nyquist criterion*

- 2-D DFT pairs between signal $f(x, y)$ and $F(u, v)$
  - The image is discrete 2-D function

- **2-D DFT properties**
  - *convolution theorem*

$$f(x, y) * h(x, y) \Leftrightarrow F(u, v)H(u, v)$$

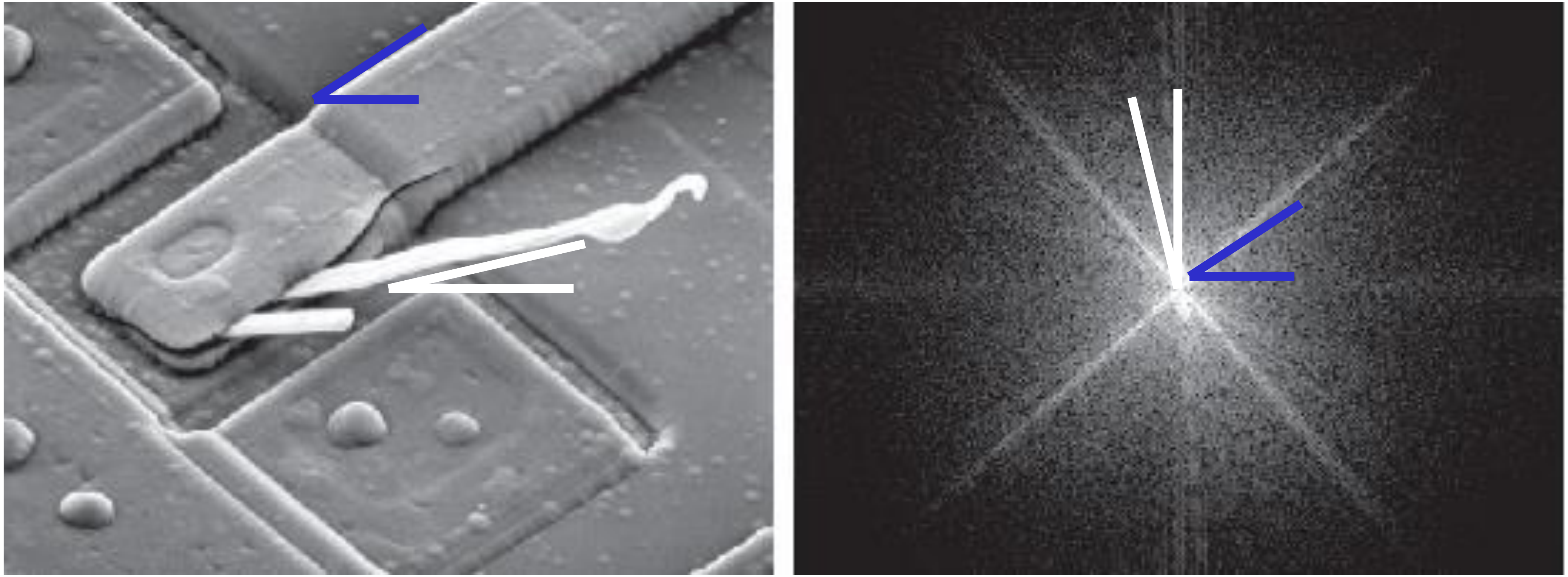
$$f(x, y)h(x, y) \Leftrightarrow F(u, v) * H(u, v)$$

- *Wavelet transform*

Contents

Spatial and Frequency

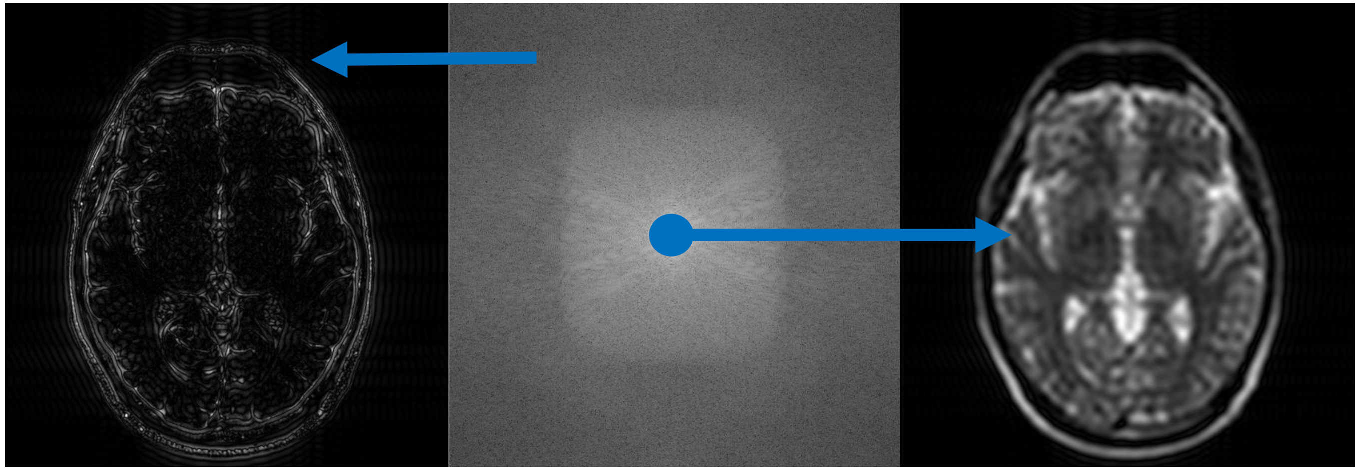
- What do the frequency components stand for?
- How could image be enhanced in the frequency domain?



SEM image of a damaged integrated circuit

Spatial and Frequency

- Low frequencies
  - Slowly varying intensity components
  - DC component
- Frequency domain
  - Sharp transitions in intensity
  - Edges and noise



Periphery of K-space *Center of K-space*

Spatial and Frequency

- Filters: Fourier transform pair

$$h(x,y) \Leftrightarrow H(u,v)$$

$h(x,y)$: finite impulse response (FIR) filters.

$$f(x,y) * h(x,y) \Leftrightarrow F(u,v)H(u,v)$$

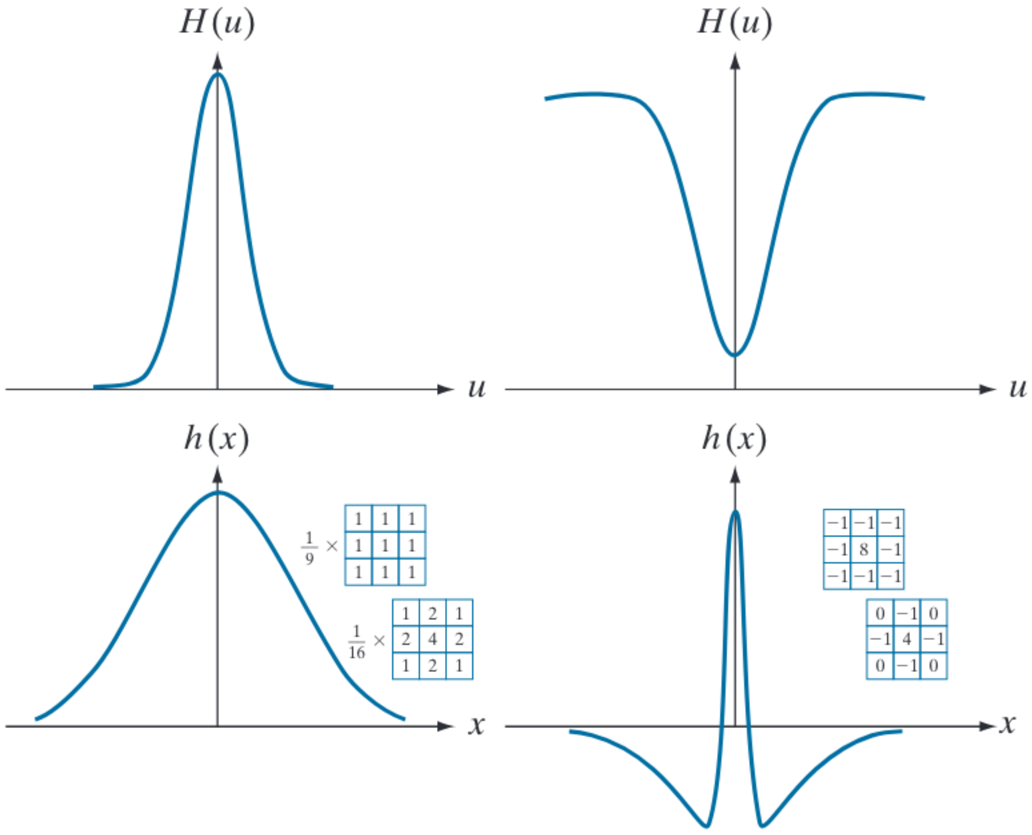
- Circular convolution

$$(f * h)(x,y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n)h(x-m, y-n)$$

Spatial and Frequency

Gaussian Frequency domain

Gaussian Spatial domain



Low pass

High pass

Narrow frequency filter
Wider spatial filter

$$f(ax, by) \Leftrightarrow \frac{1}{|ab|} F(\frac{u}{a}, \frac{u}{b})$$

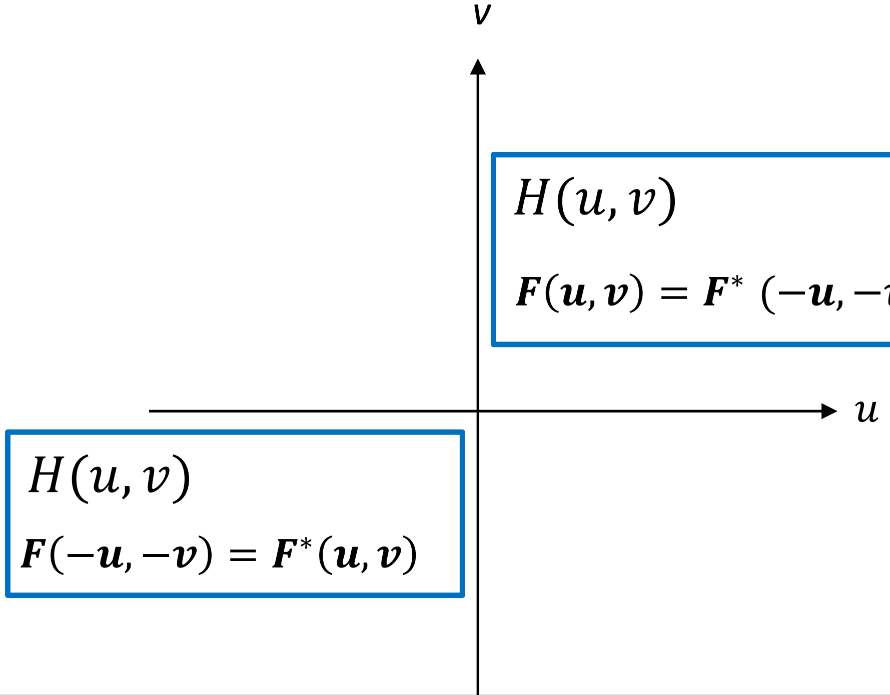
Fundamentals

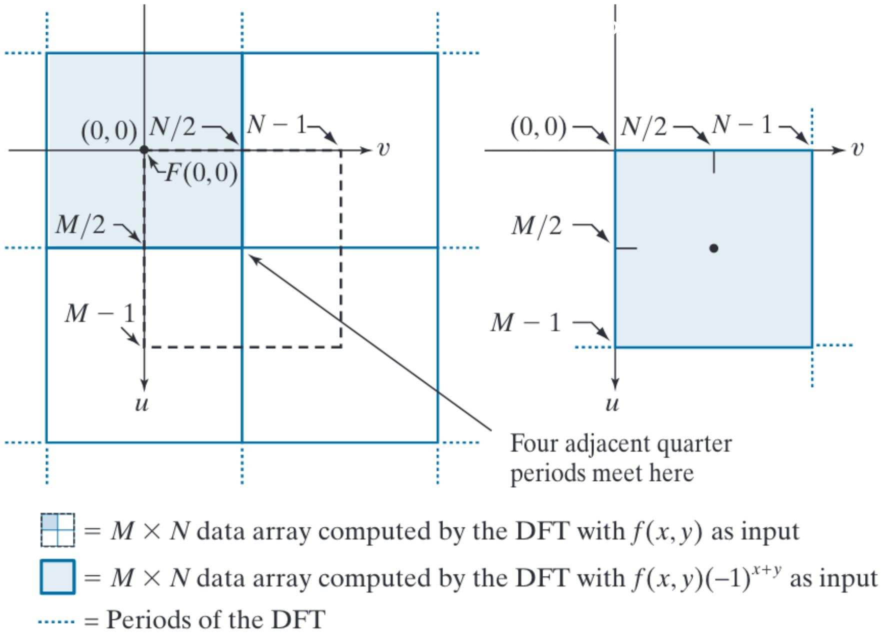
Basic filtering equation

$$g(x,y) = Real\{\mathcal{F}^{-1}[H(u,v)F(u,v)]\}$$

Output image *Filter (transfer function)* *DFT of input image f(x,y)*

- Filter is simplified if it's symmetric about the center
- $(-1)^{x+y}$ prior to computing (or *fftshift* after DFT)





Fundamentals

- Filtering in the frequency domain

1. Fourier transfer an image

$$\mathcal{F}[f(x,y)] = F(u,v)$$

2. Modify its transform

$$\boldsymbol{H(u,v)}F(u,v)$$

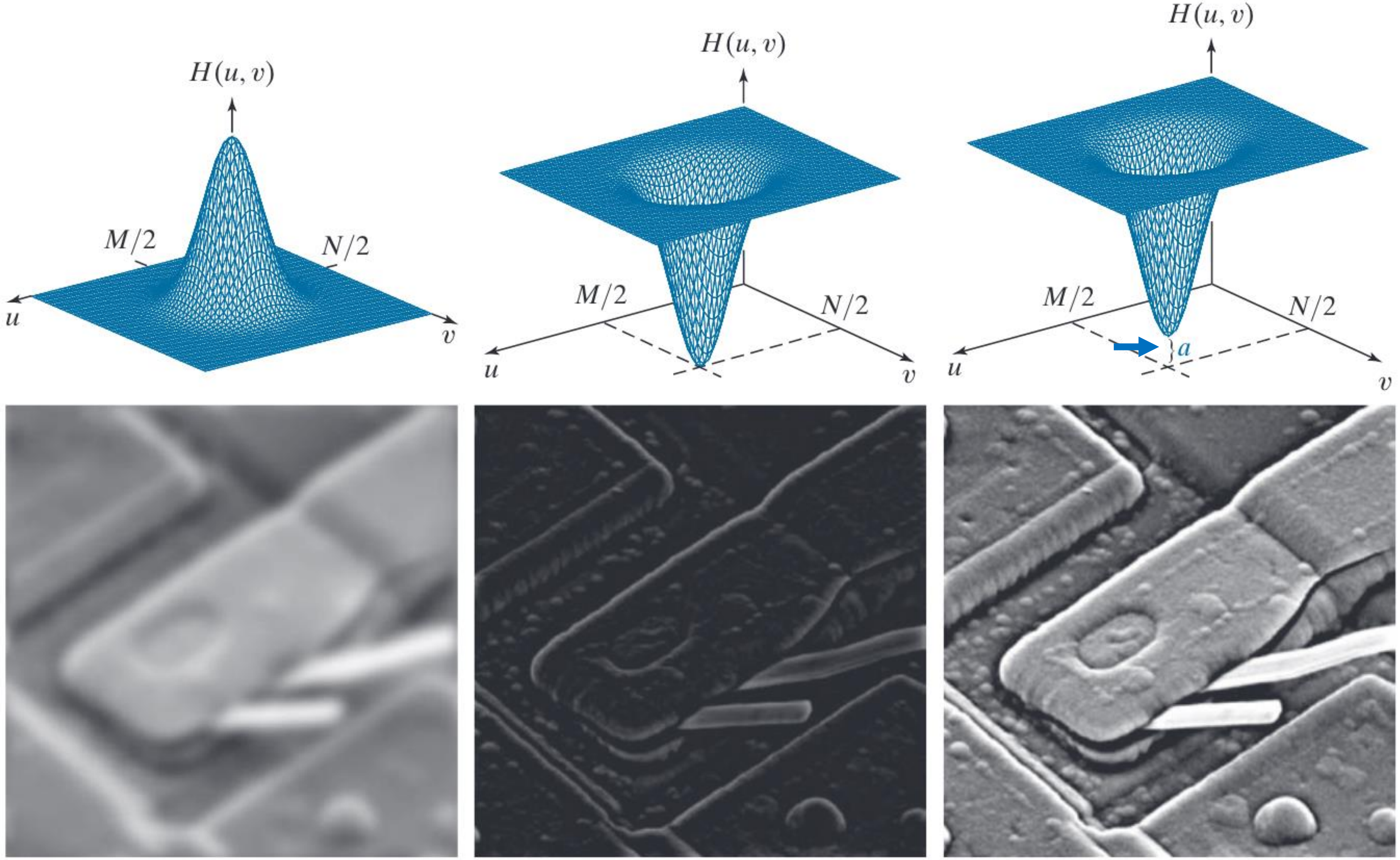
3. Inverse transform back to an image

$$\mathcal{F}^{-1}[H(u,v)F(u,v)]$$

4. Assume real image

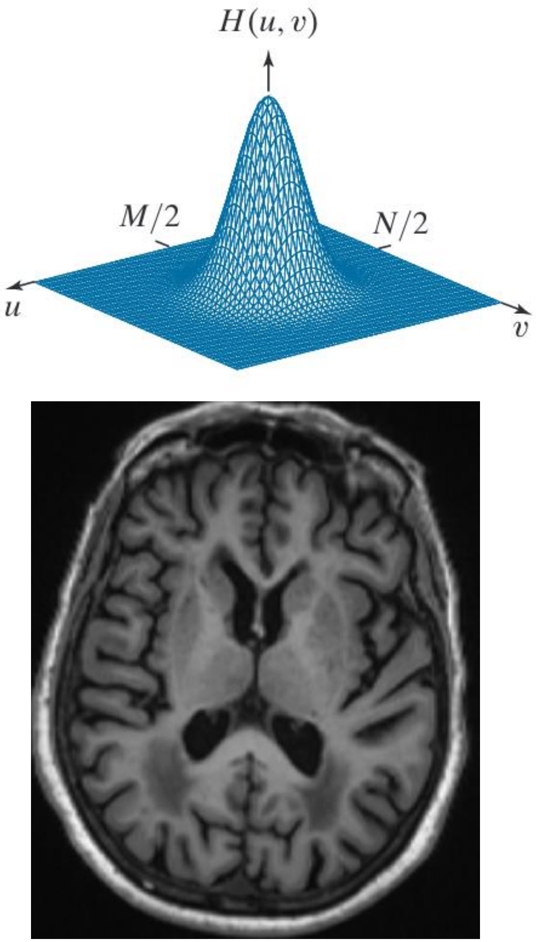
$$g(x,y) = real\{\mathcal{F}^{-1}[H(u,v)F(u,v)]\}$$

Fundamentals

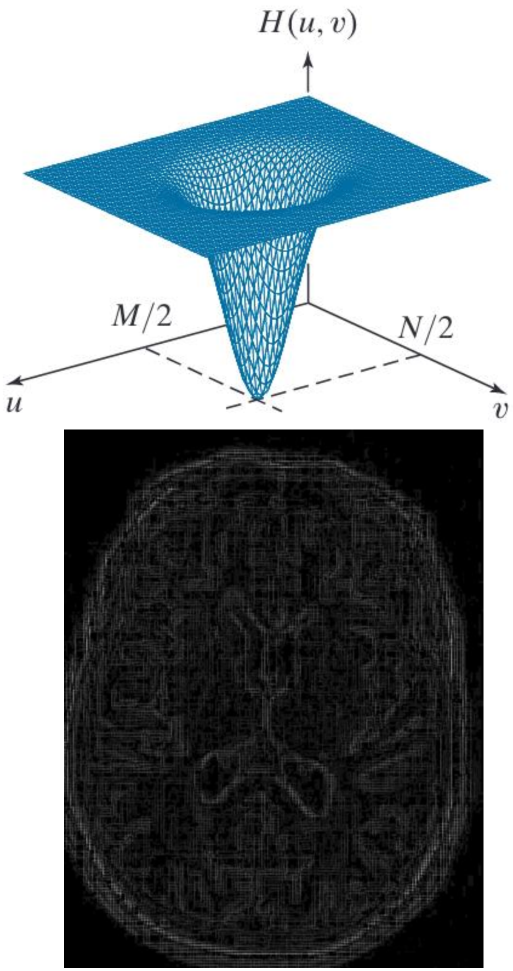


Low pass High pass Offseted high pass

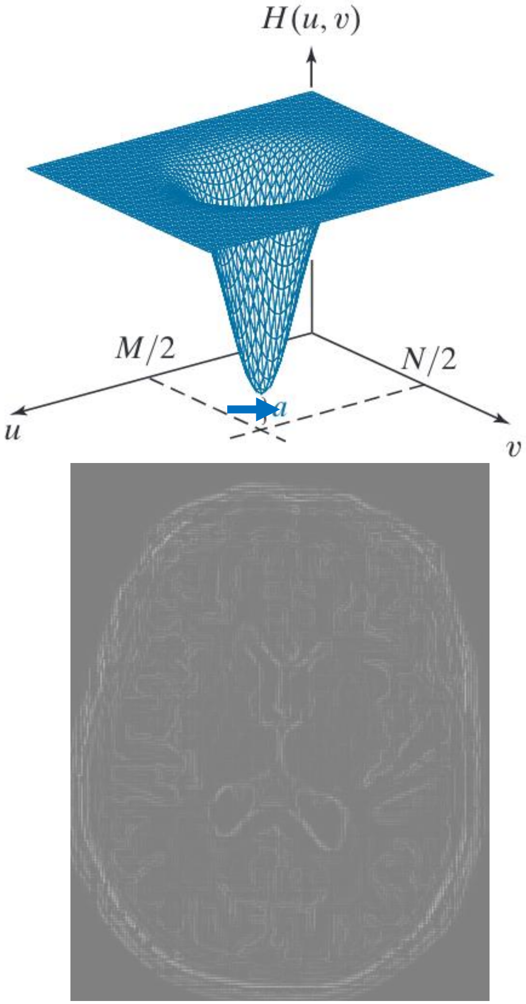
Fundamentals



Low pass



High pass



Offseted high pass

Fundamentals

- Spatial convolution: wraparound error
  - if the functions in question are not padded.

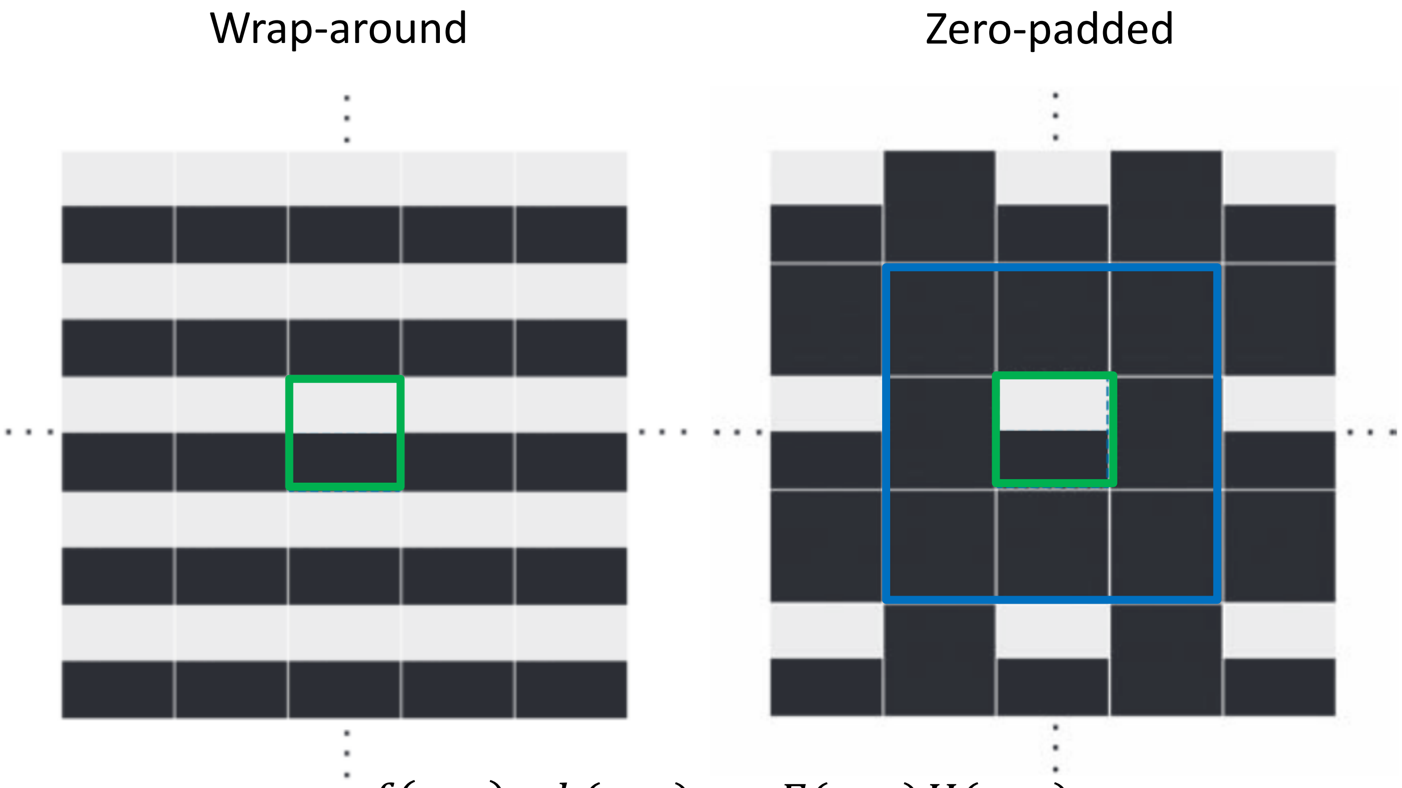
Gaussian low pass

Without padding

Zero-padding

Fundamentals

- Spatial convolution: wraparound error
  - if the functions in question are not padded.

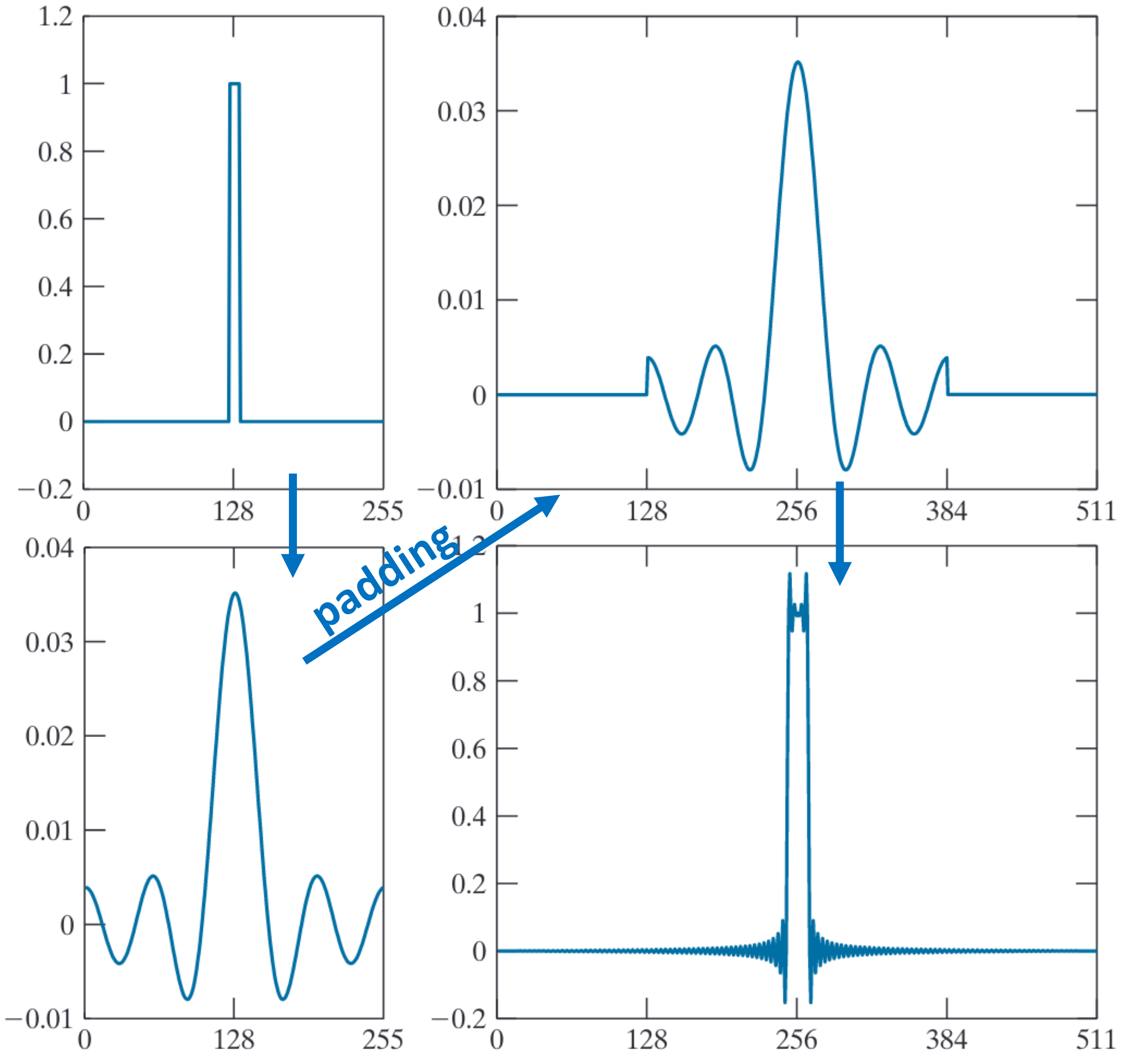


$$f(x,y) * h(x,y) \Leftrightarrow F(u,v)H(u,v)$$

Fundamentals

frequency delta

Spatial Sinc

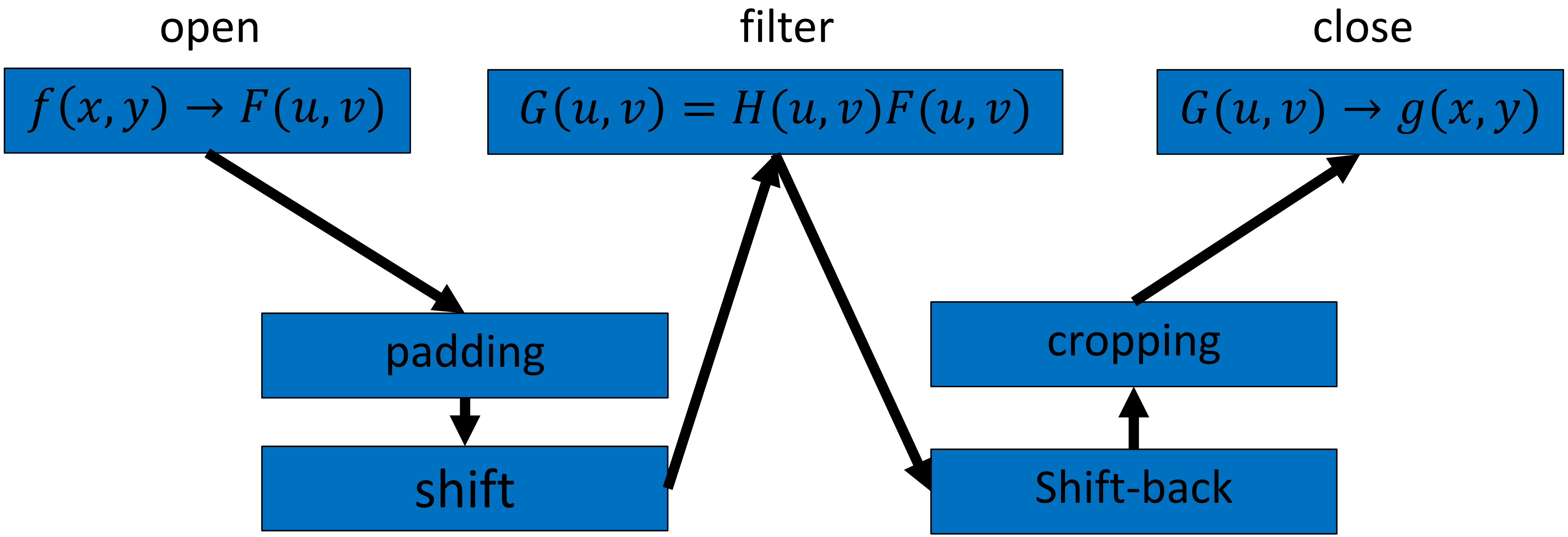


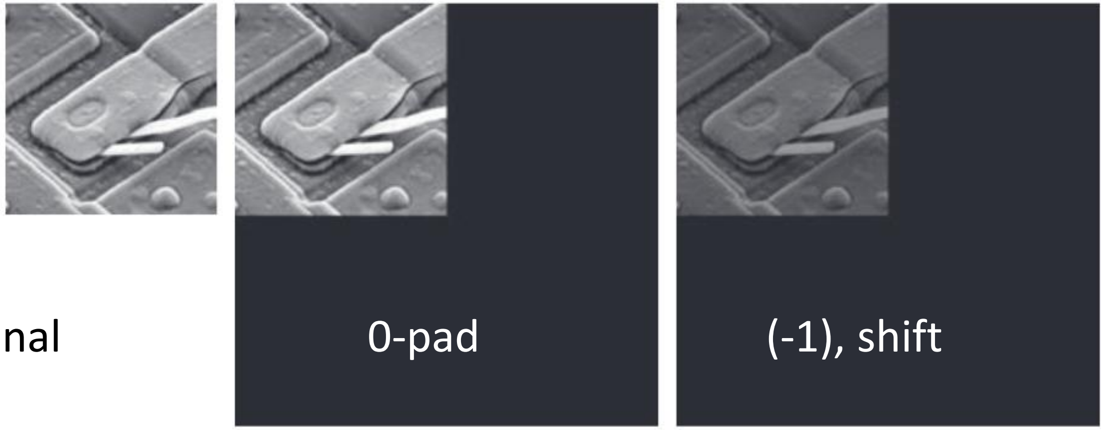
Spatial Sinc
Zero-padding

Frequency delta
Gibbs-ring

How to put an elephant in fridge?

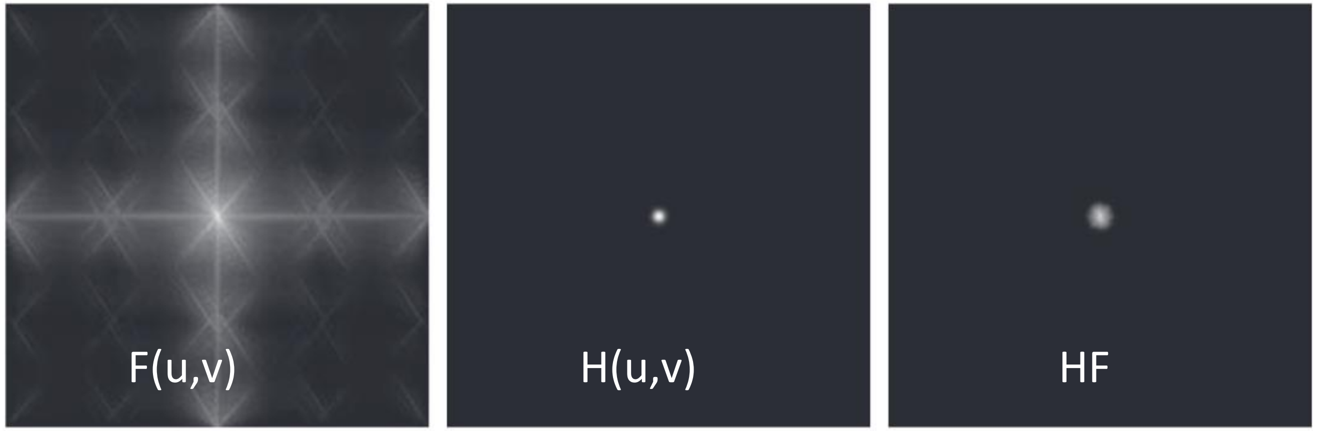
Spatial and Frequency





original

Multiply $f_p(x,y)$ by -1^{x+y}



$$g_{p(x,y)} = \left(real\left[F^{-1}\left(G(u,v)\right)\right]\right)(-1)^{x+y}$$

Steps of frequency filtering

- For an image $f(x, y)$ of size $M \times N$, padding to $P \times Q$

$$P = 2M, Q = 2N$$

- Multiply $f_p(x, y)$ by -1^{x+y} to center the Fourier transform
- DFT: $F(u, v)$
- Filter function $H(u, v)$ with size $P \times Q$, centered at $(P/2, Q/2)$

$$G(u, v) = H(u, v)F(u, v)$$

- iDFT

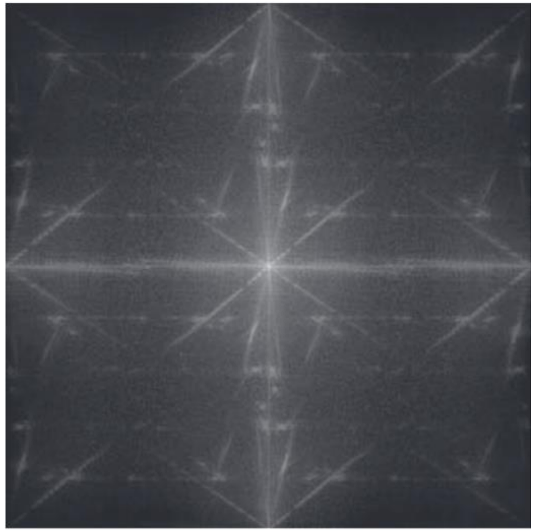
$$g_{p(x,y)} = \left(real\left[F^{-1}\left(G(u, v)\right)\right]\right)(-1)^{x+y}$$

- Crop $M \times N$

Spatial and frequency filtering

image

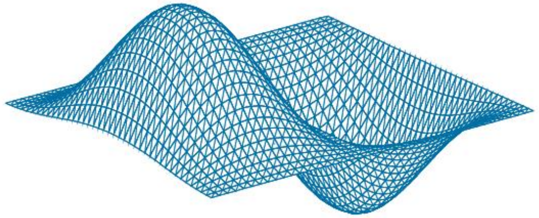
spectrum



−1	0	1
−2	0	2
−1	0	1

$h(x, y)$

$H(u, v)$

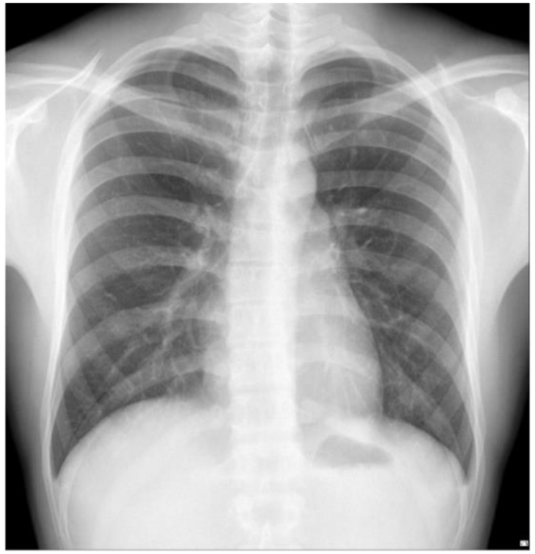


Spatial filtering

Frequency filtering

Spatial and frequency filtering

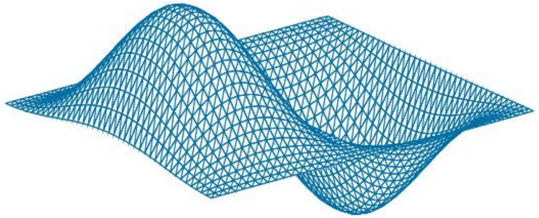
image

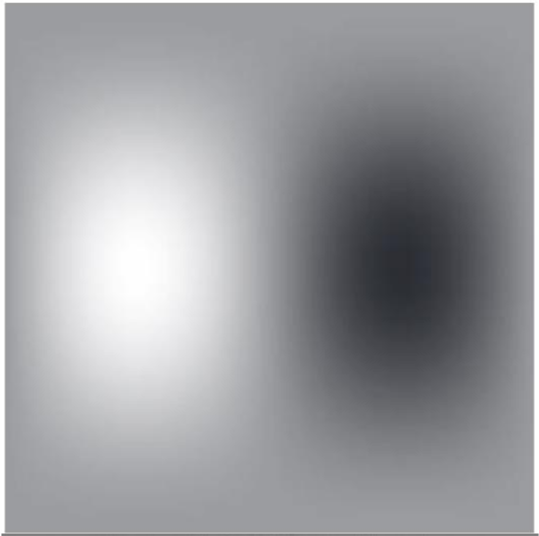


−1	0	1
−2	0	2
−1	0	1

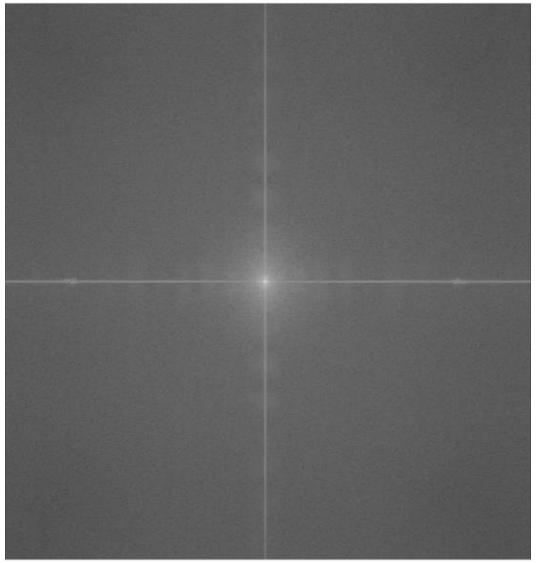
$h(x, y)$

$H(u, v)$





spectrum



Spatial filtering

Frequency filtering

Summary

- Spatial kernels to frequency filters

$$h(x,y) \Leftrightarrow H(u,v)$$

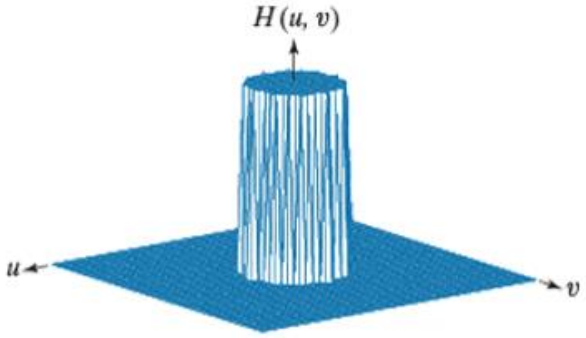
- DFT spectrum
  - Magnitude is usually used
- Basic filtering equation

$$g(x,y) = Real\{\mathcal{F}^{-1}[H(u,v)F(u,v)]\}$$

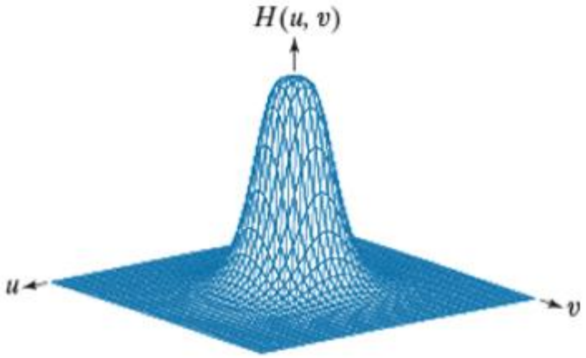
- Steps for filtering
  - Multiply $(-1)^{x+y}$ (fftshift)
  - padding
- Open→filter→close

Frequency domain: Lowpass

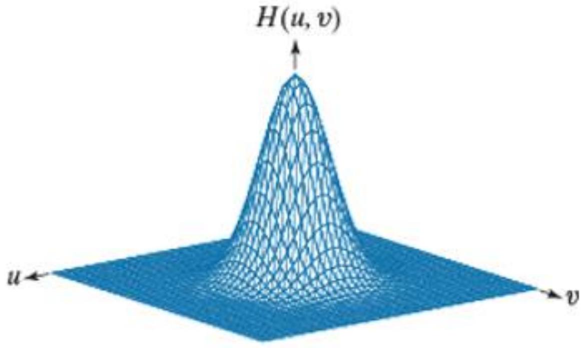
- lowpass (smoothing) filters:
  - Ideal
  - Butterworth (nth order)
  - Gaussian



Ideal



Butterworth

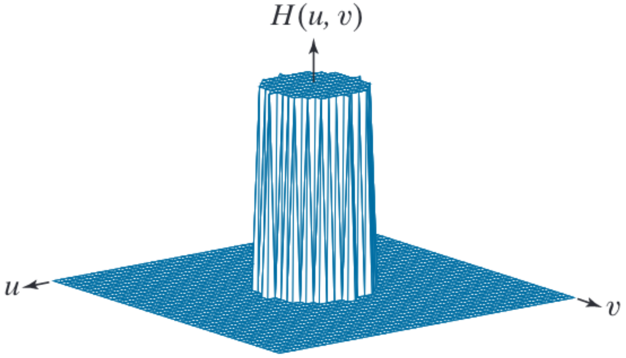


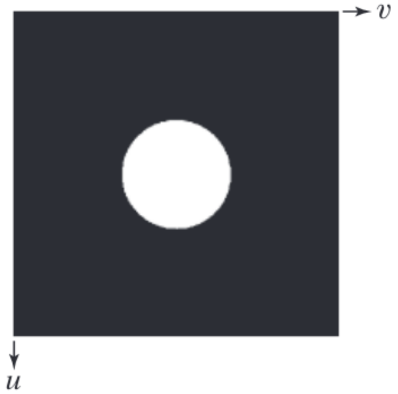
Gaussian

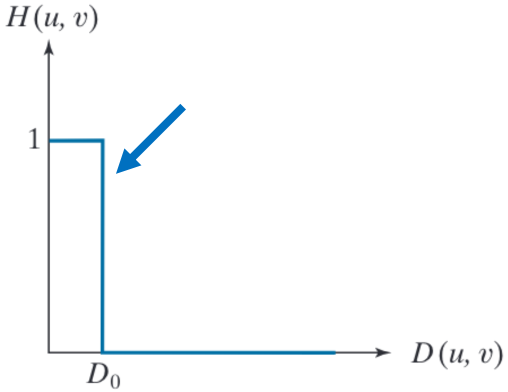
Ideal filter

- Frequency

$$H(u,v) = \begin{cases} 1, & if\ \mathrm{D(u,v)} \le D_0 \\ 0, & if\ \mathrm{D(u,v)} > D_0 \end{cases} \qquad D(u,v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$



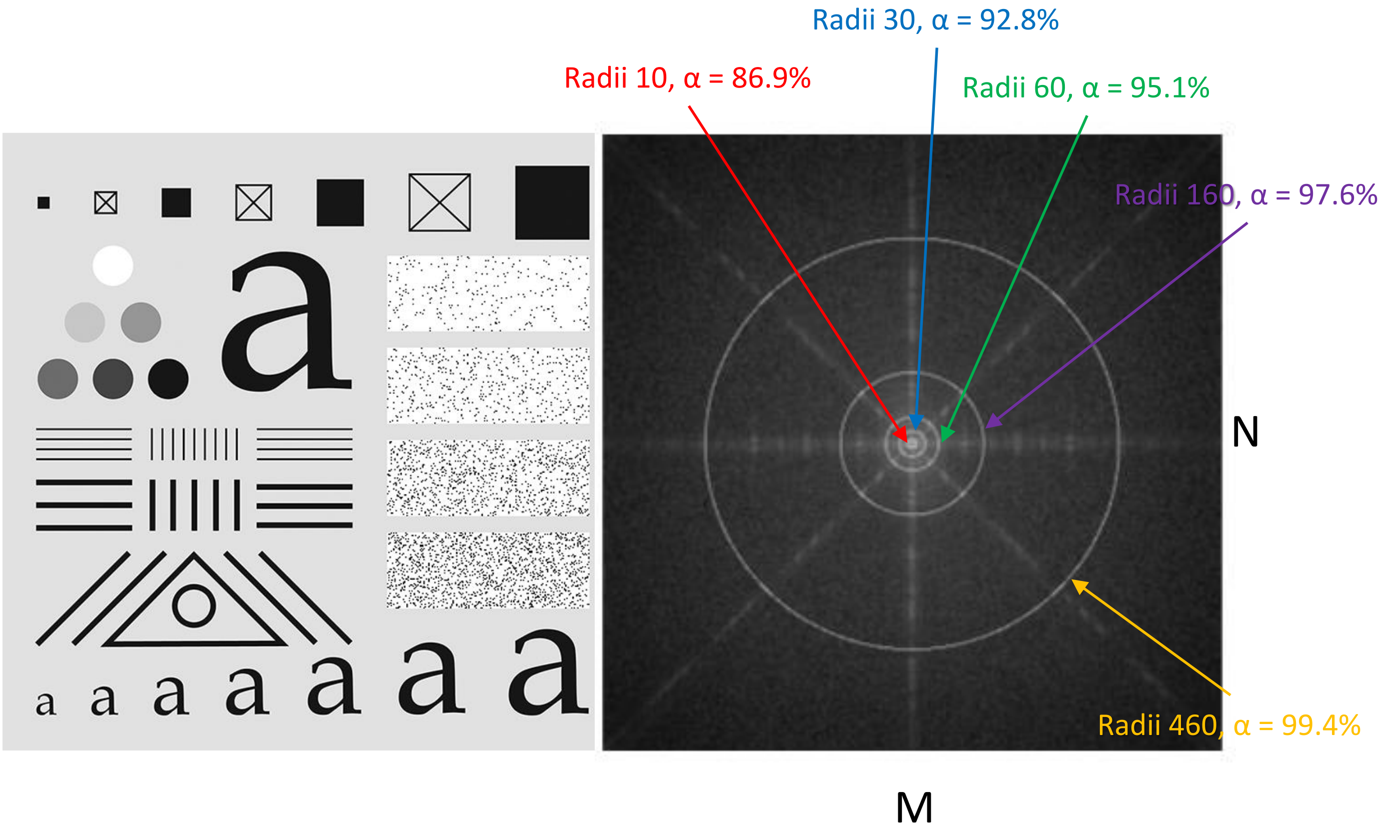




cutoff

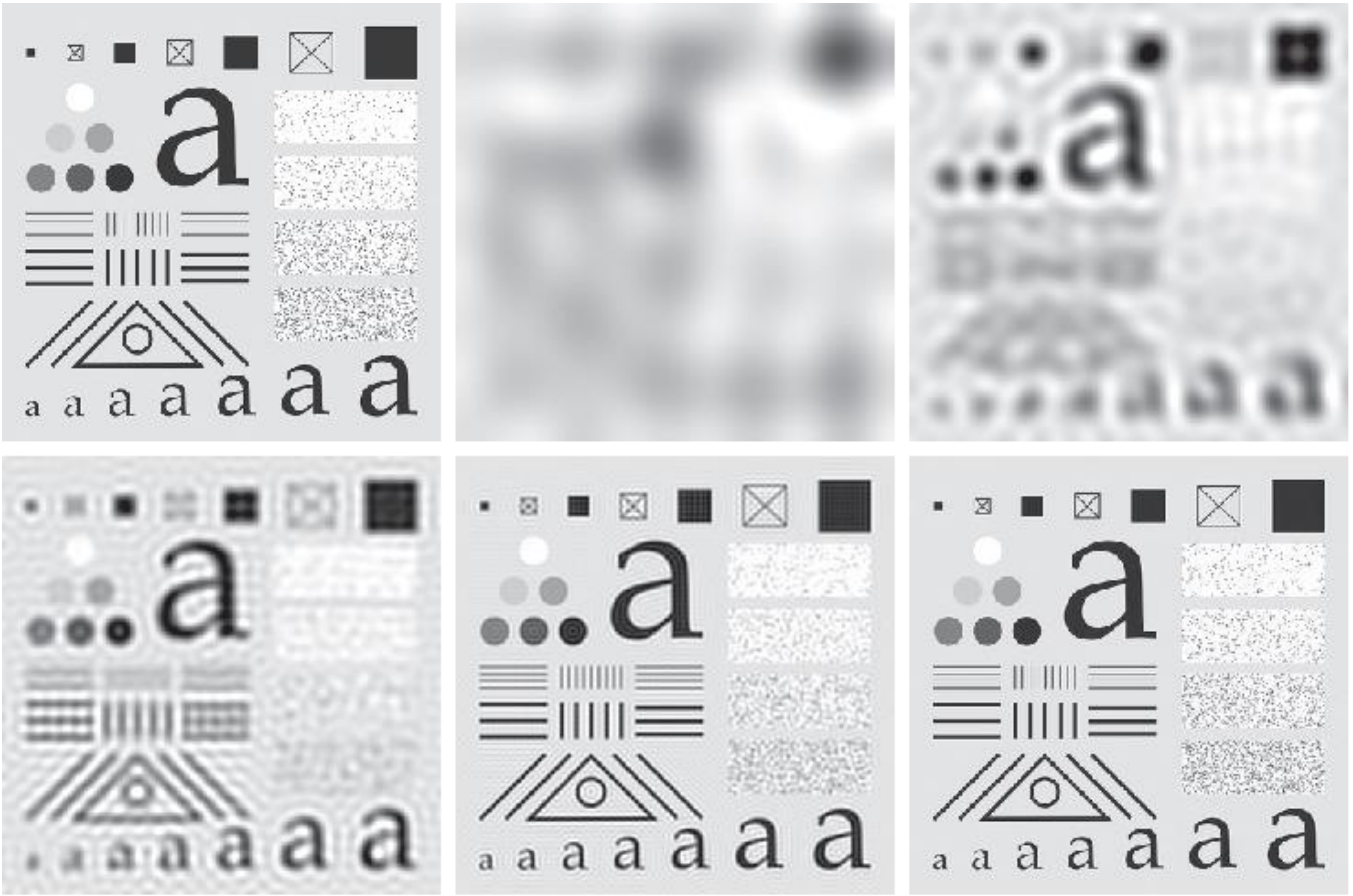
- Ideal: passed without attenuation

Ideal filter



The radii enclose 86.9, 92.8, 95.1, 97.6, and 99.4% of the padded image power

Ideal filter



“Ringing” effects
More blur, more ringing!

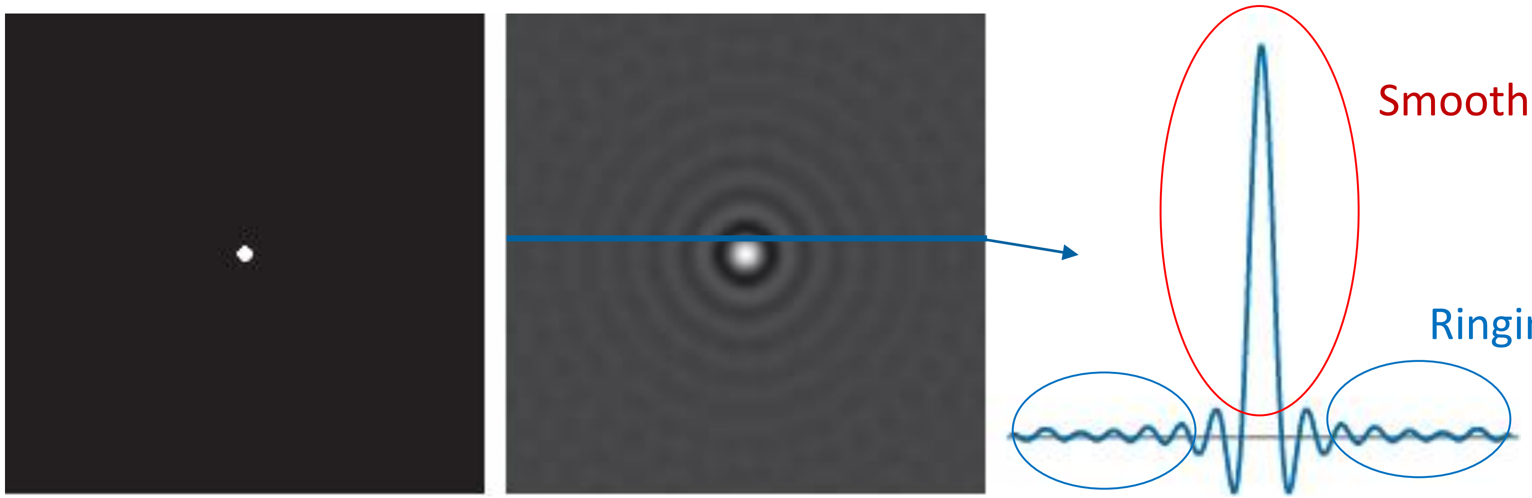
Cutoff frequencies set at radii 10,30,60,160, and 460 (original 688 x 688)

Ideal filter

Ideal filter

ILPF function in frequency domain

Intensity profile



Corresponding spatial domain kernel function

The reciprocal nature between $H(u, v)$ and $h(x, y)$, along with the convolution process, explains mathematically why the blurring and ringing are more severe the narrower the filter in the frequency domain

Frequency domain: Lowpass

Is there any filter can smooth an image without ringing effects?

Basic idea:

Make the falling edge of the ideal low pass filter smoother in frequency domain

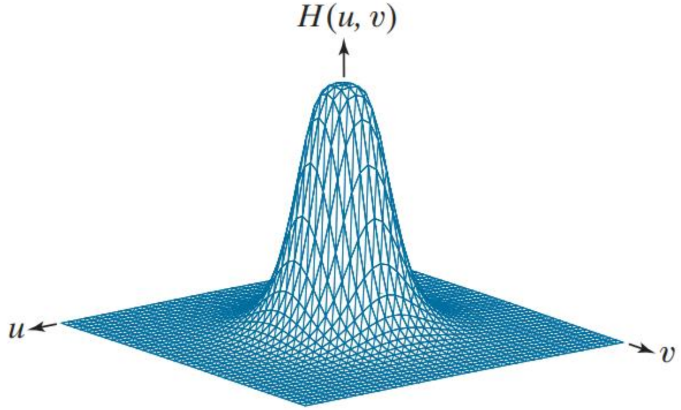
Butterworth

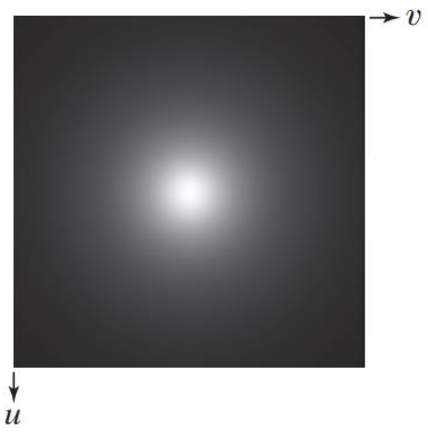
Gaussian

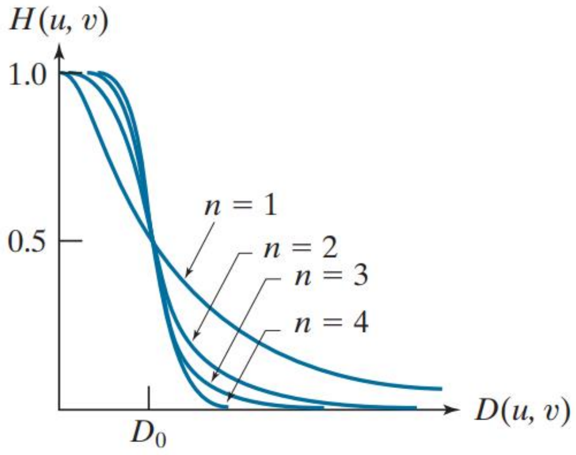
Butterworth

$$H(u,v) = \frac{1}{1 + [D\,(u,v)/D_0]^{2n}}$$

$$D(u,v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

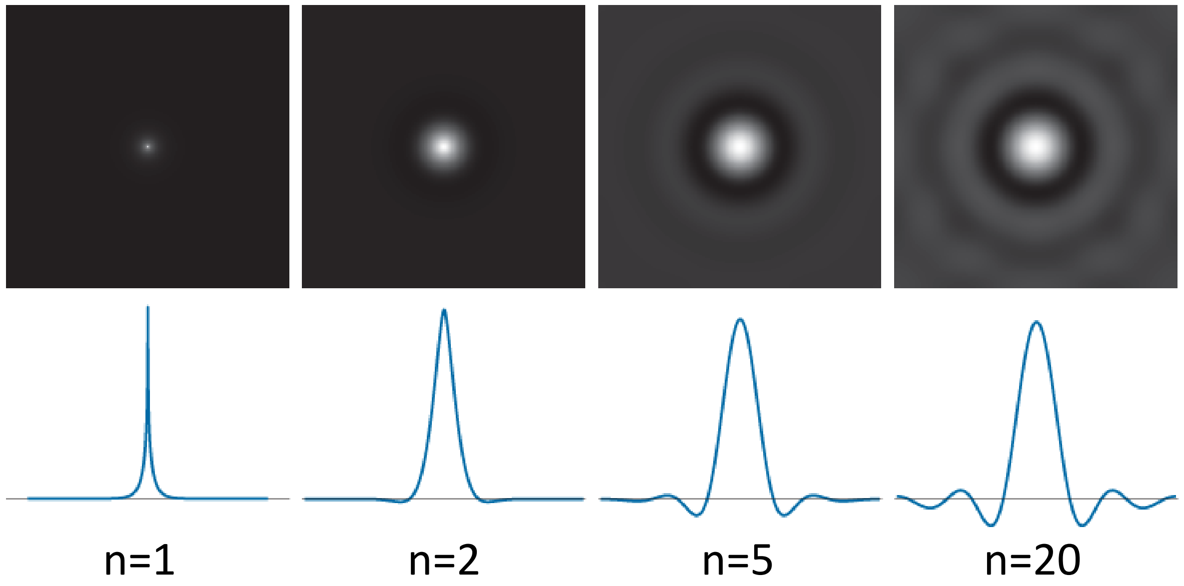






Butterworth

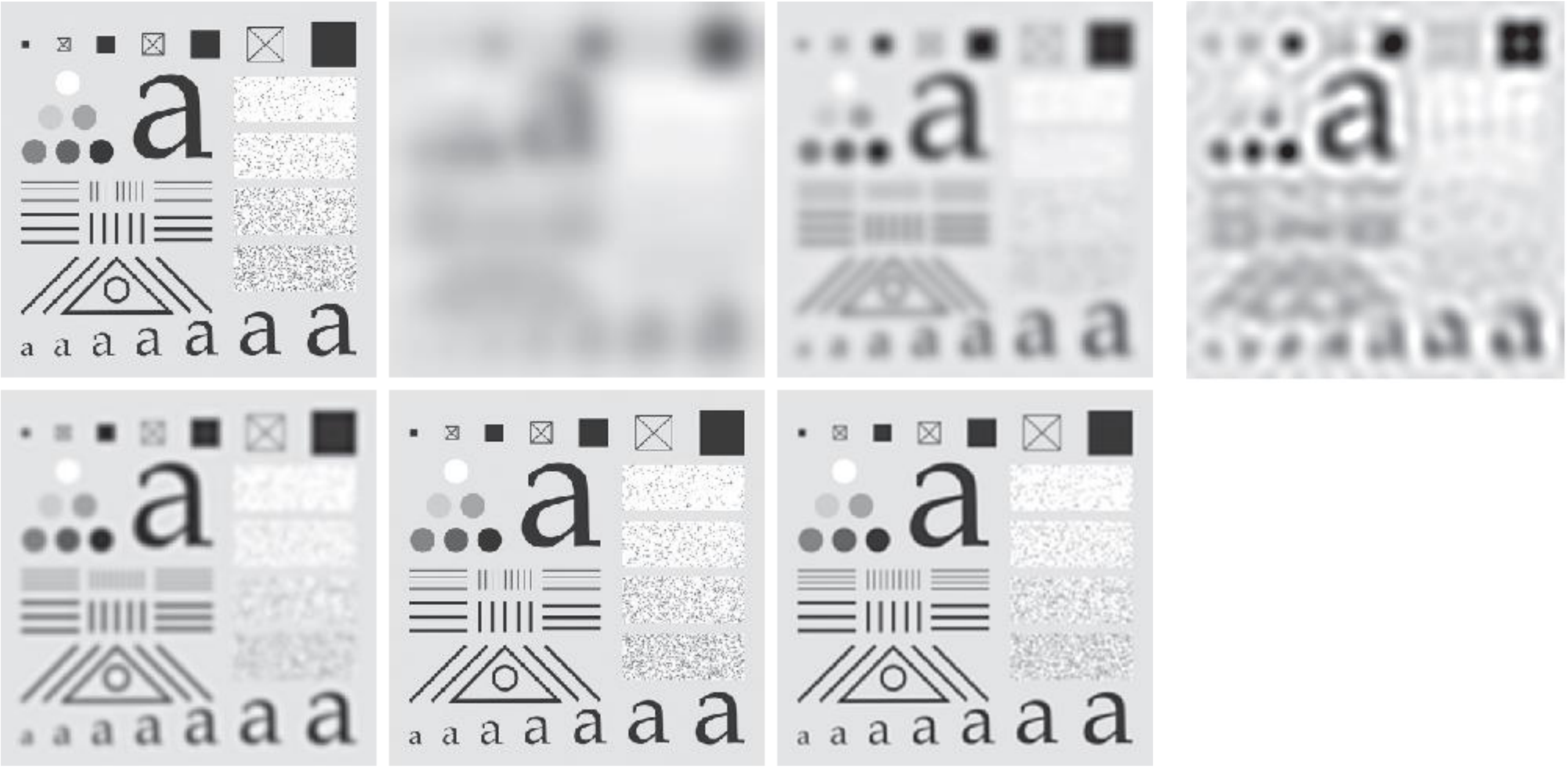
- imperceptible in filters of order 2 or 3
- significant in filters of higher orders



Butterworth

No visible ringing

Idea filter



D_0 of 10,30,60,160, and 460 pixels with n=2.25

Gaussian Filter

C.F. Gauss

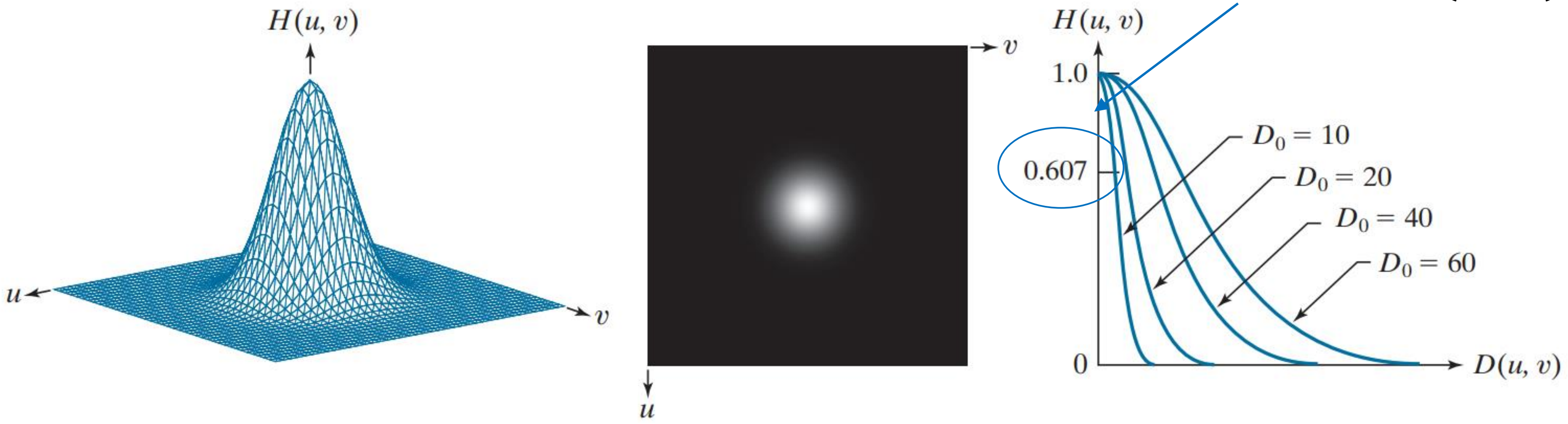
Magical Gaussian function:

- Many events in nature are subject to Gaussian distribution, such as height, performance, noise
- The derivative of a Gaussian is still a Gaussian
- The Fourier transform of the Gaussian is still the Gaussian
- The inverse Fourier transform of a Gaussian is still a Gaussian

Gaussian Filter

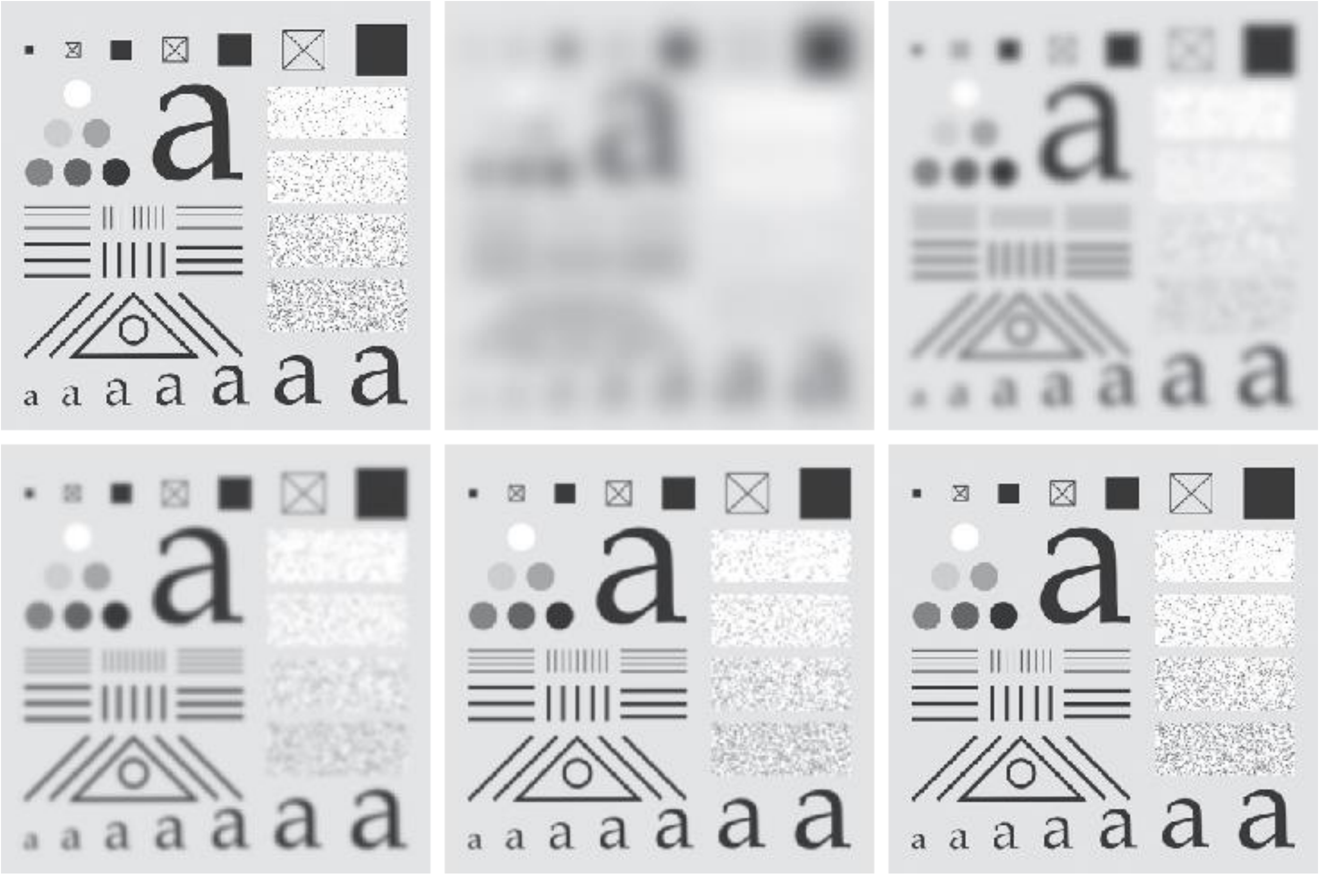
$$H(u,v) = e^{-D^2(u,v)/2\sigma^2} \qquad H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

when $D(u,v) = D_0$



Gaussian Filter

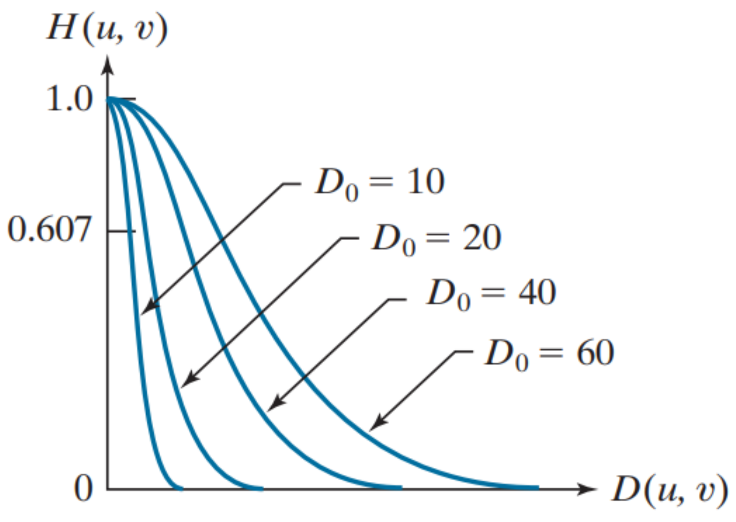
$D_0 = 10$ $D_0 = 30$

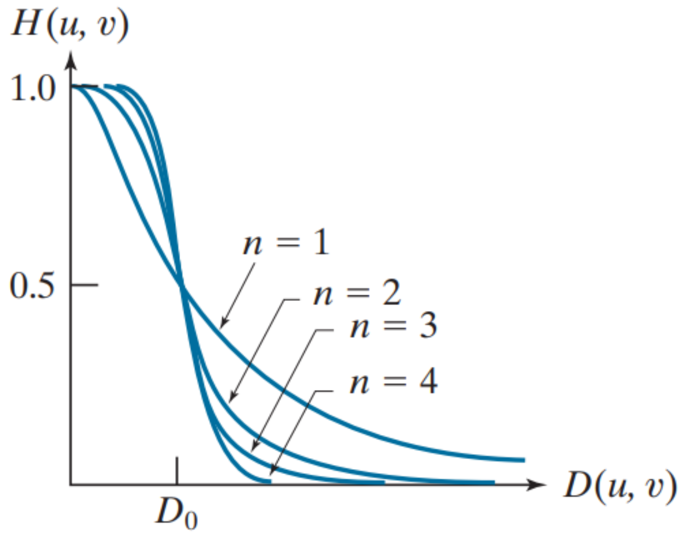


$D_0 = 60$ $D_0 = 160$ $D_0 = 460$

No ringing at all!

Gaussian Lowpass Filters V.S. Butterworth Lowpass Filters

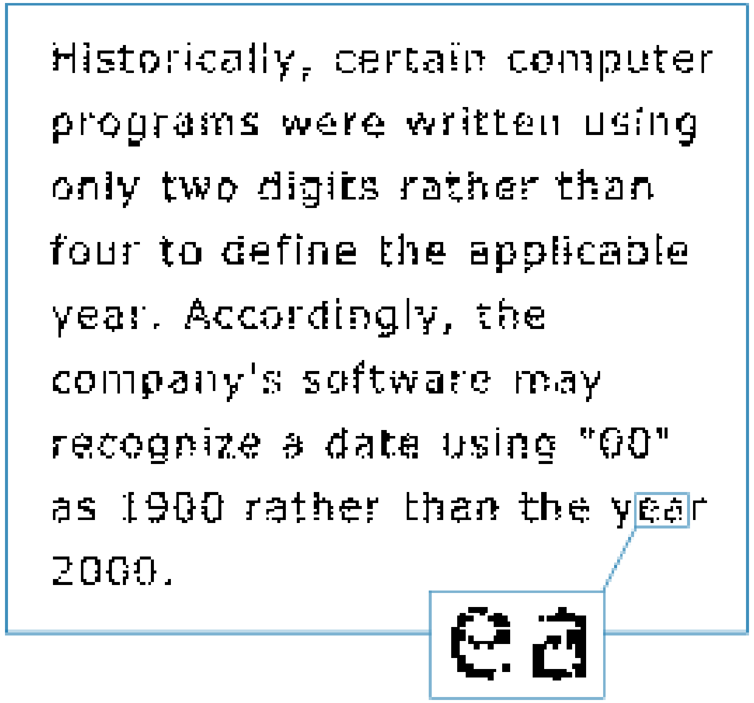




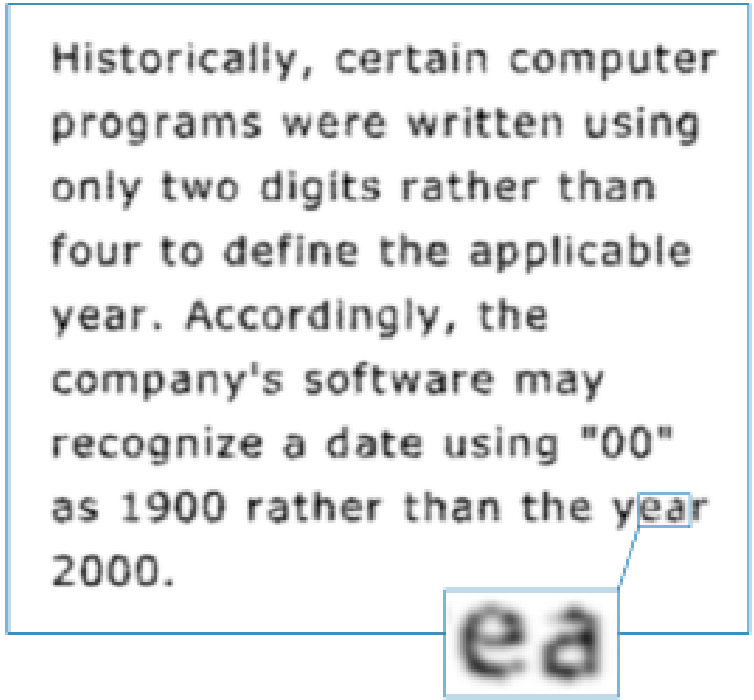
1. Gaussian can completely eliminate ringing while Butterworth can't
2. At the cutoff frequency, Gaussian is not as sharp as Butterworth

- When tight control of the transition between low and high frequencies about the cutoff frequency — **Butterworth**
- When no ringing is allowed (e.g. in Medical diagnosis) — **Gaussian**

Examples

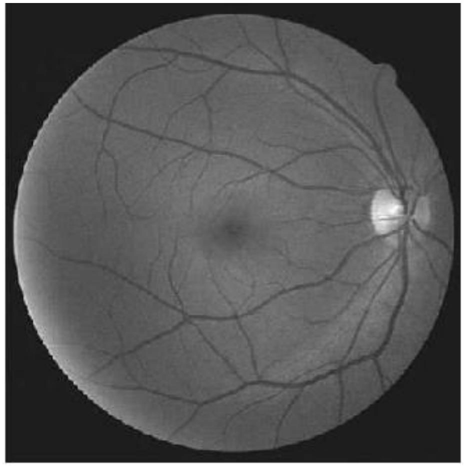

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

Broke charactors

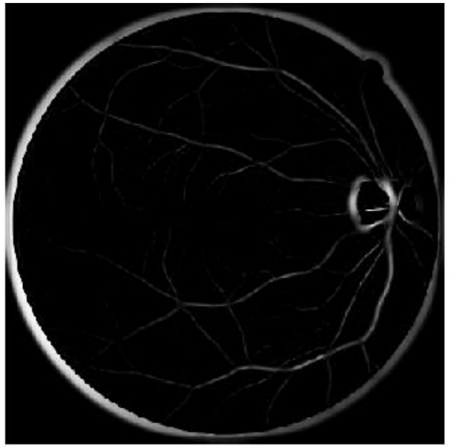

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

repair

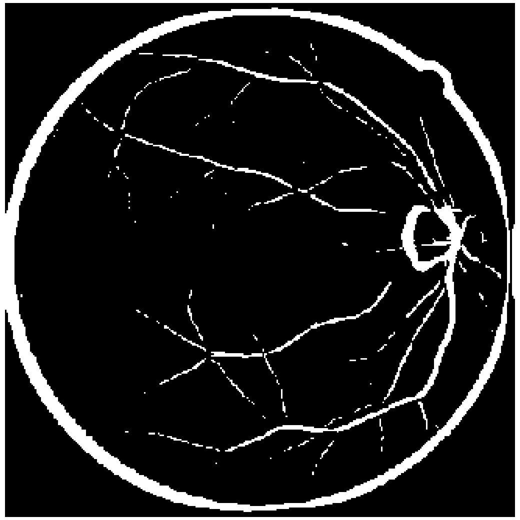
Examples



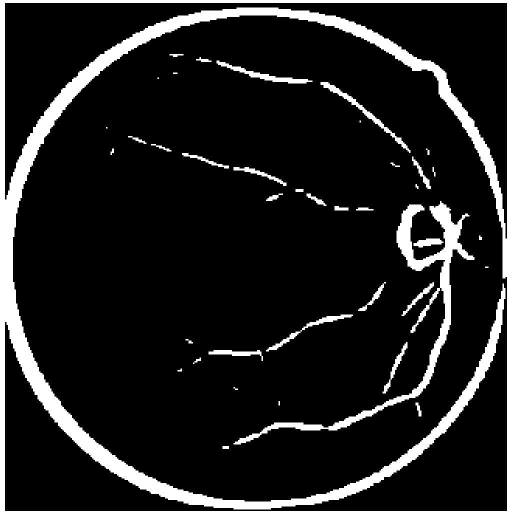
retinal



enhanced vessel



sharpness



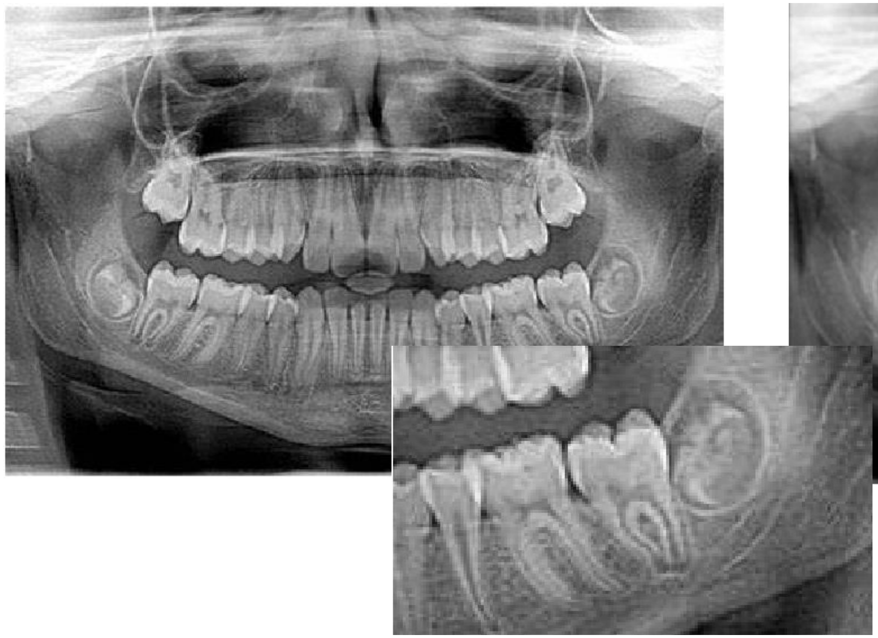
main vessel

Reduce the sharpness of major vessel and small vessels

Examples

reduce the sharpness of fine skin lines and small blemishes

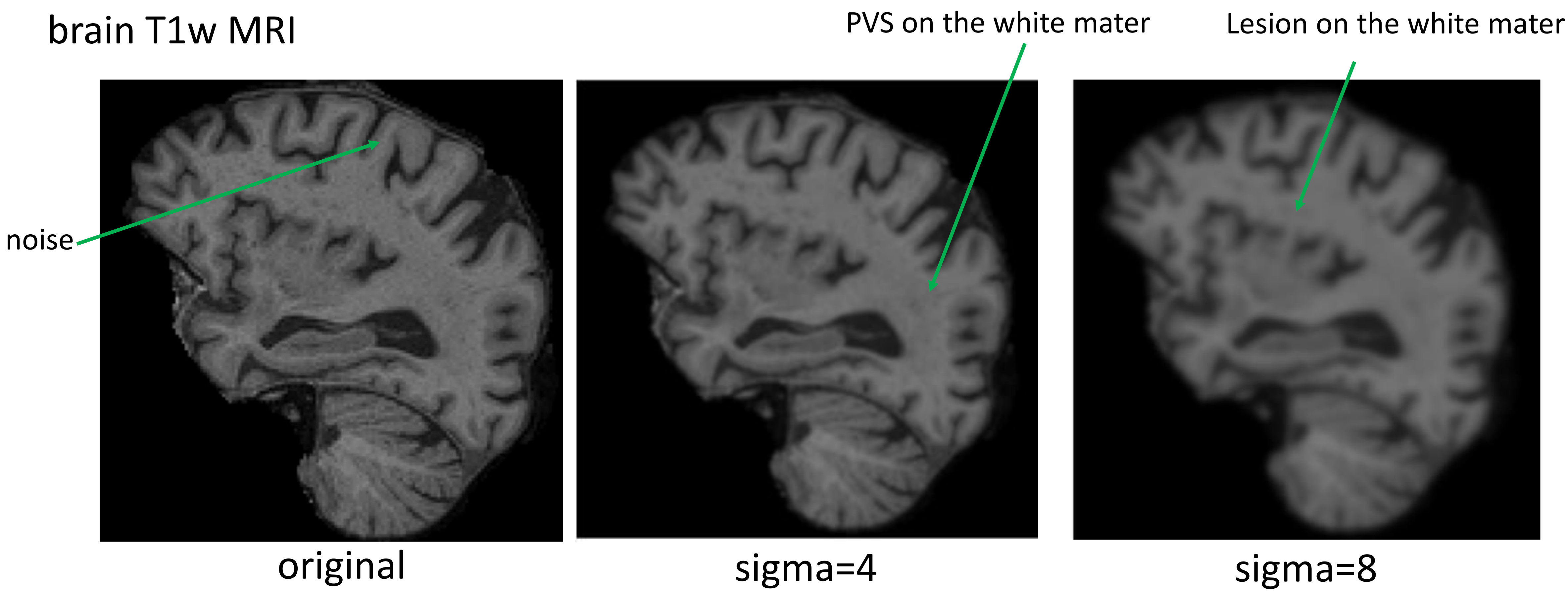
Examples



Reduce the noise

Examples

brain T1w MRI



Frequency domain: Highpass

- Conceptually, the intended function of the filters is to perform precisely the reverse operation of the ideal lowpass filters:
- Image sharpening can be achieved in the frequency domain by a highpass filtering process.

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

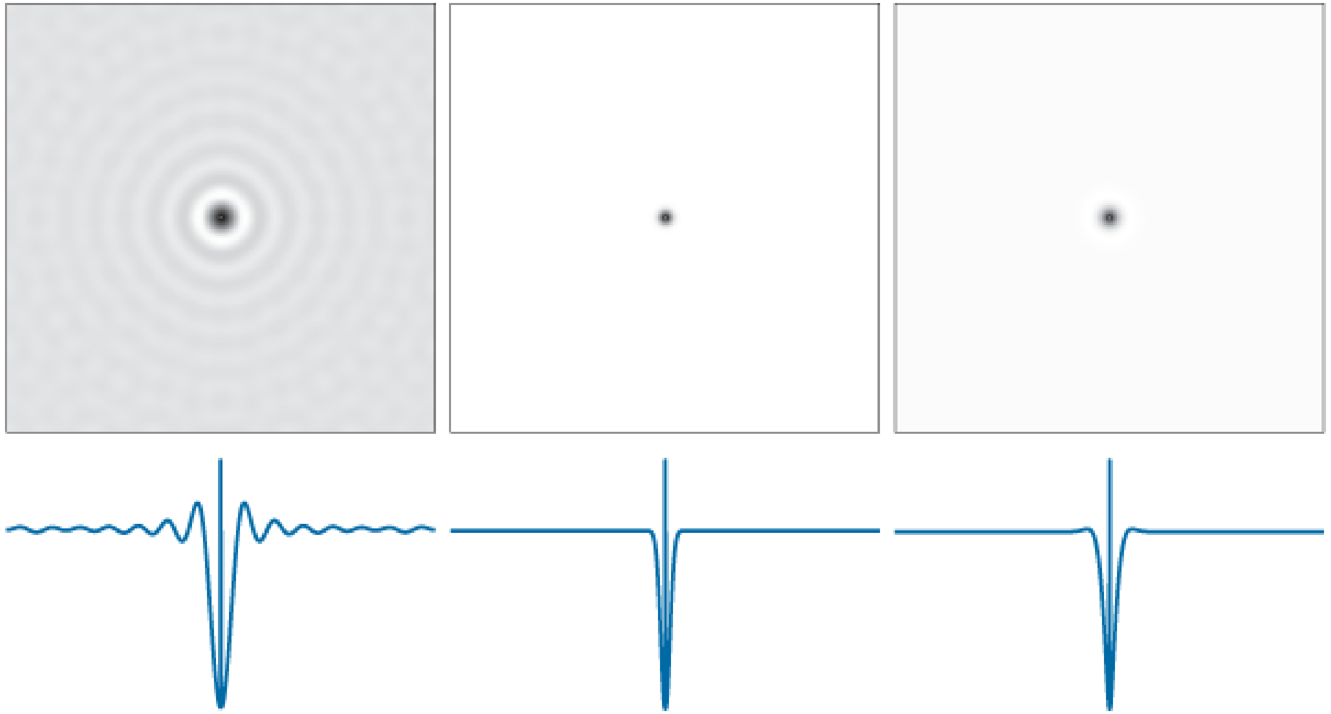
Ideal

Butterworth

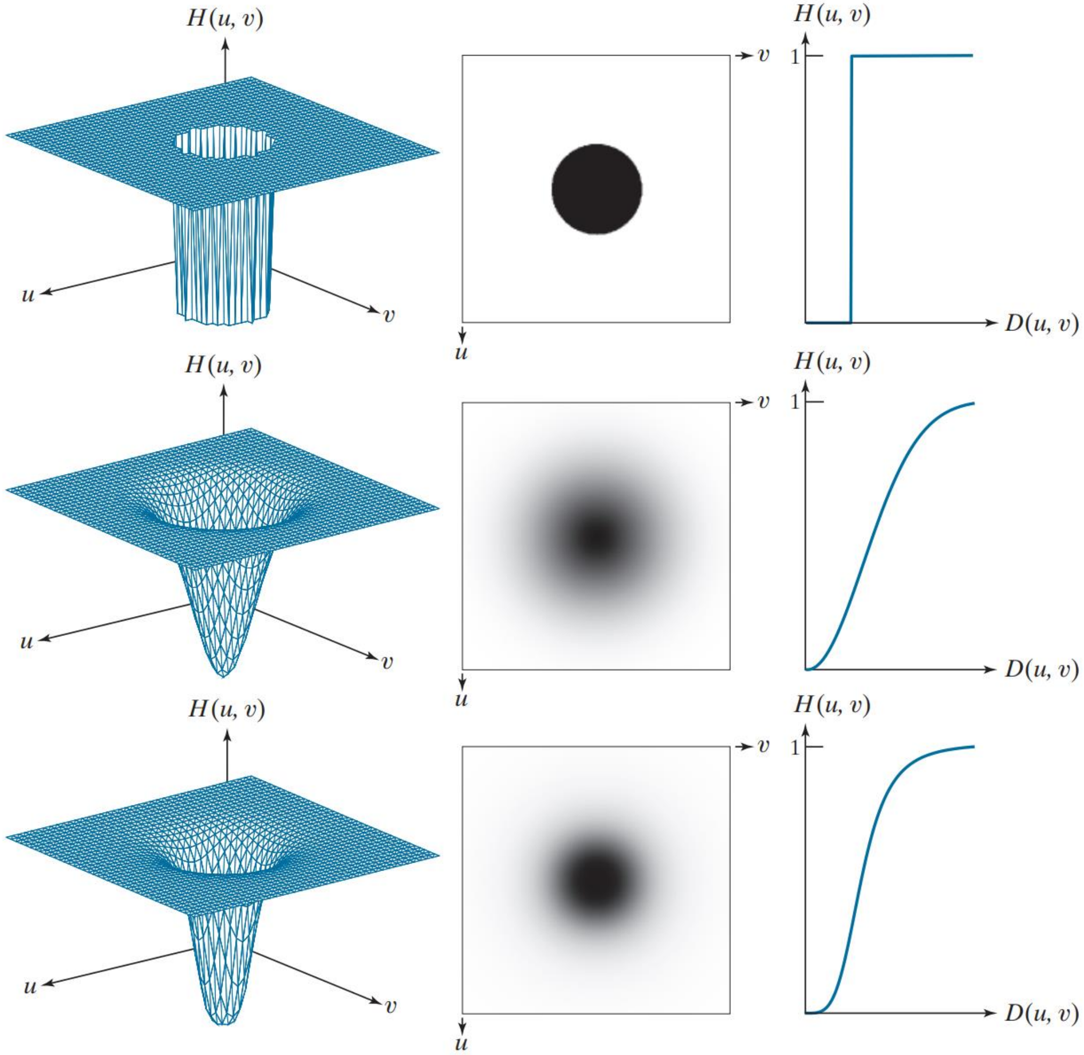
Gaussian

Frequency domain: Highpass

$$
\begin{aligned}
h_{HP(x,y)} &= \mathfrak{I}^{-1}\left[H_{HP(u,v)}\right] \\
&= \mathfrak{I}^{-1}\left[1 - H_{LP(u,v)}\right] \\
&= \delta(x,y) - h_{LP(x,y)}
\end{aligned}
$$



Frequency domain: Highpass



Ideal

$$H(u,v) = \begin{cases} 0, & if\ \mathrm{D(u,v)} \le D_0 \\ 1, & if\ \mathrm{D(u,v)} > D_0 \end{cases}$$

Gaussian

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

Butterworth

$$H(u,v) = 1 - \frac{1}{1 + [D\,(u,v)/D_0]^{2n}}$$

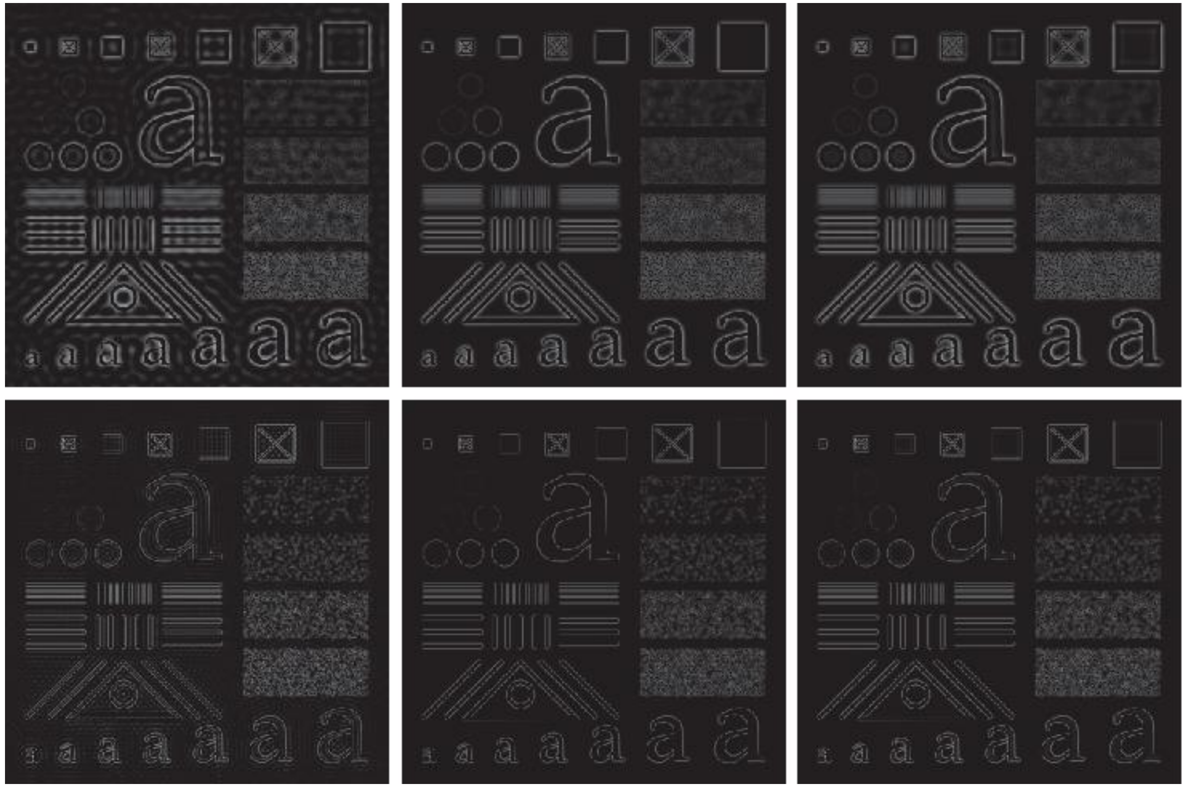
Examples

Ringing! Sharp transit

IHPF GHPF BHPF

D=60

D=160

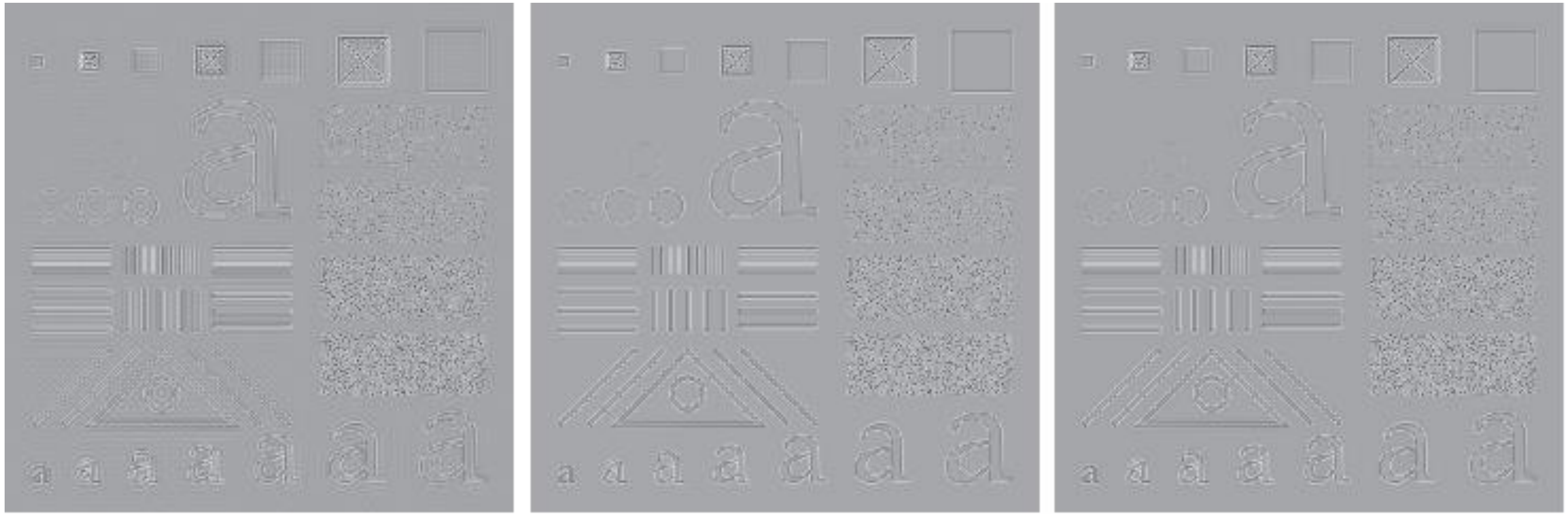


Clear boundary

negative values are clipped by the display at 0 (black) DC term to zero

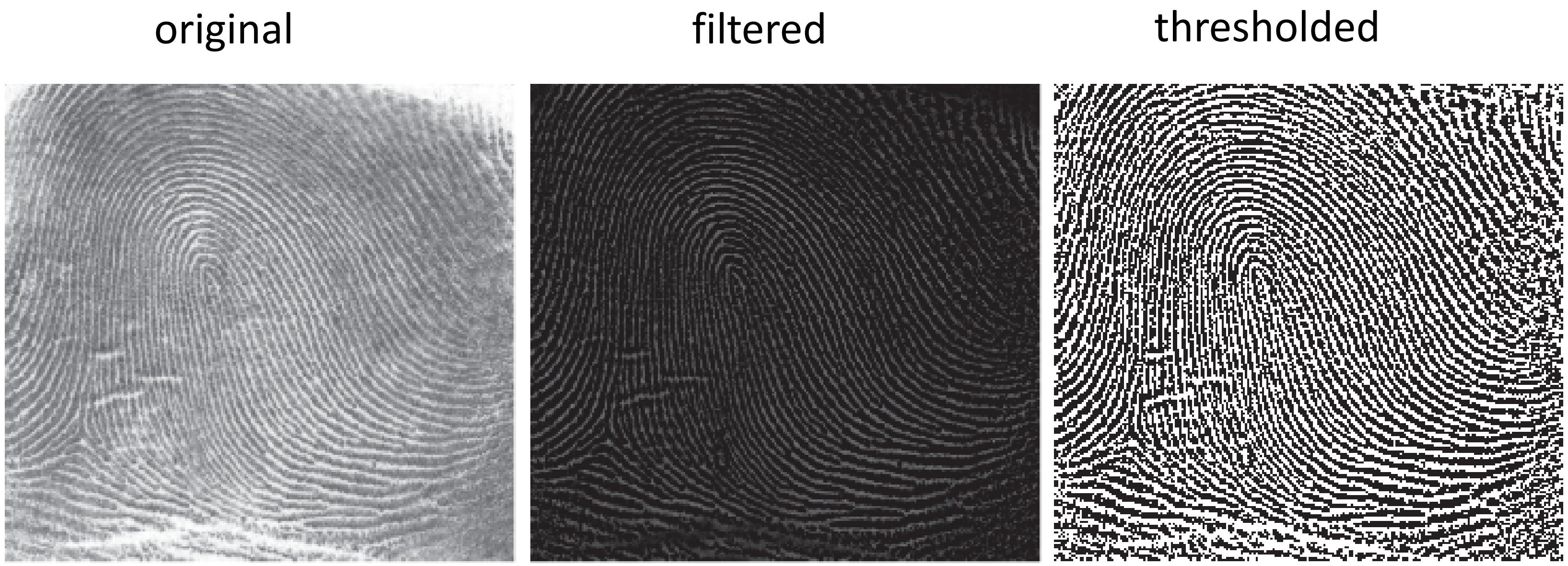
Examples

IHPF GHPF BHPF



positive and negative intensities

Examples



Butterworth highpass filter of order 4 with a cutoff frequency of 50.

Frequency domain: Laplacian filter

$$\Im\left[\frac{\partial^2 f(x,y)}{dx^2} + \frac{\partial^2 f(x,y)}{dy^2}\right] = (j2\pi u)^2 F(u,v) + (j2\pi v)^2 F(u,v)$$
$$= -4\pi^2(u^2 + v^2)F(u,v)$$

$$\Im[\nabla^2 f(x,y)] = \boxed{-4\pi^2(u^2 + v^2)}F(u,v)$$

Laplacian Filter $H(u,v)$

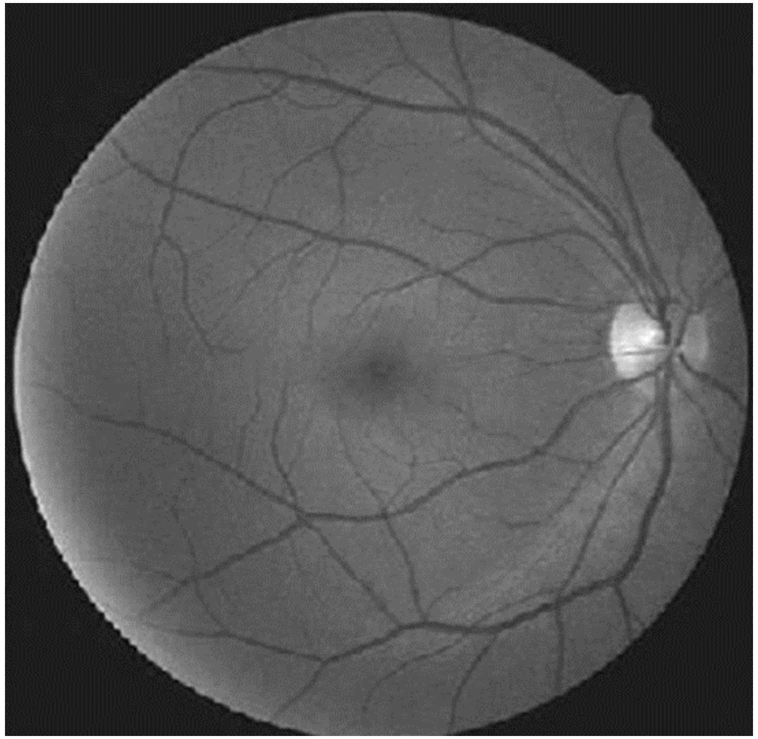
$$H(u,v) = -4\pi^2[(u - M/2)^2 + (v - N/2)^2]$$

The Laplacian-enhanced image

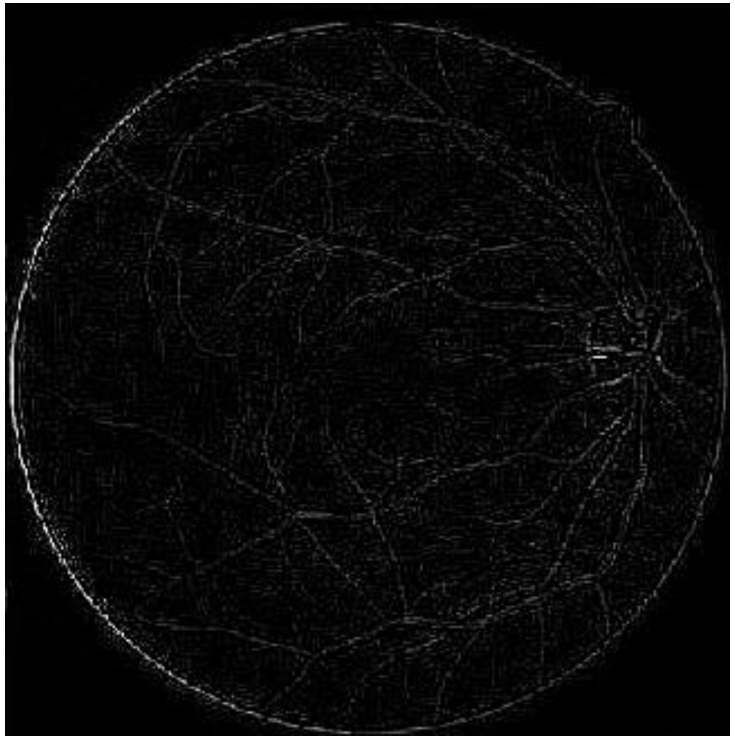
$$g(x,y) = f(x,y) + c\nabla^2 f(x,y)$$

$$g(x,y) = \Im^{-1}\{F(u,v) - H(u,v)F(u,v)\}$$
$$= \Im^{-1}\{[1 + 4\pi^2 D^2(u,v)]F(u,v)\}$$

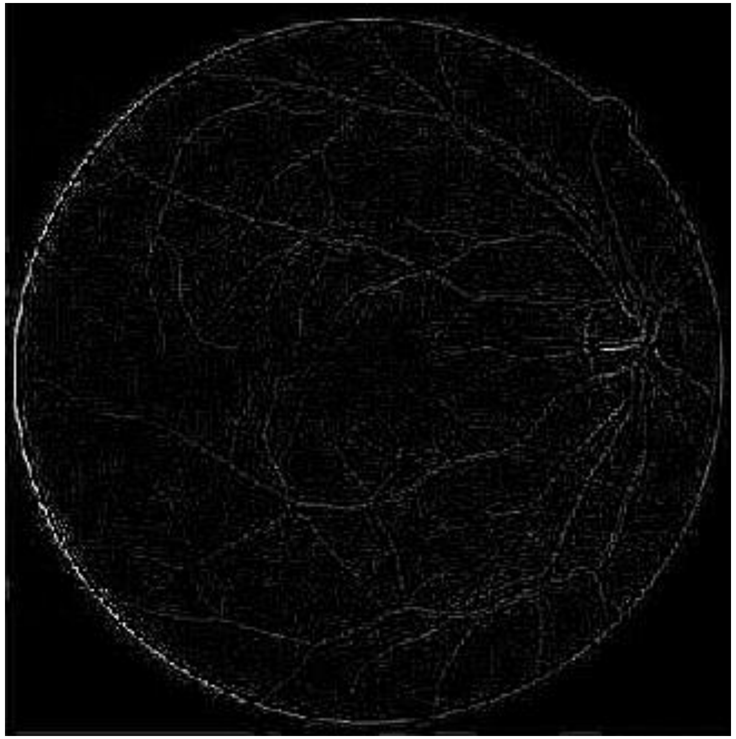
Examples



Original blurry image



Enhanced image using Laplacian (in frequency domain)



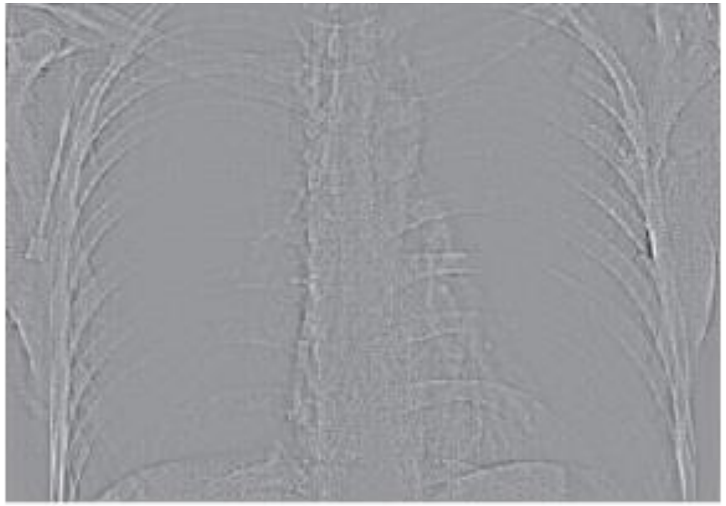
Enhanced image using Laplacian (in Spatial domain)

1	1	1
1	−8	1
1	1	1

Unsharp Masking/High-Frequency-Emphasis Filtering

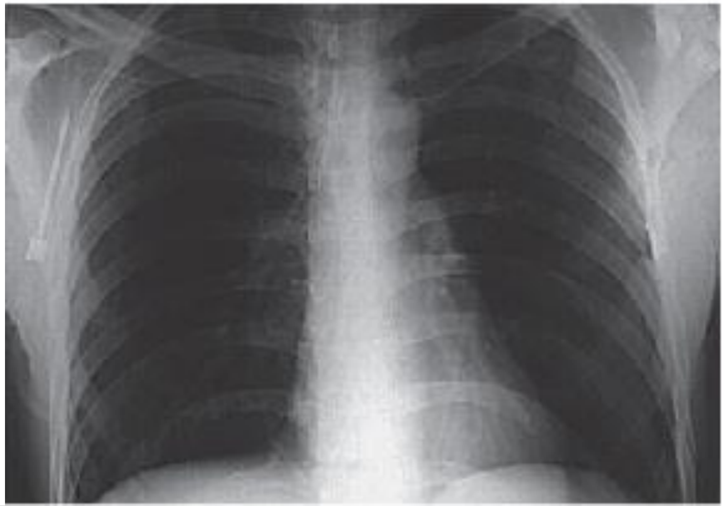
$$g(x,y) = f(x,y) + k g_{mask}(x,y) \Rightarrow g(x,y) = f(x,y) + \mathfrak{I}^{-1}\{[1 + kH_{HP}(u,v)]F(u,v)\}$$

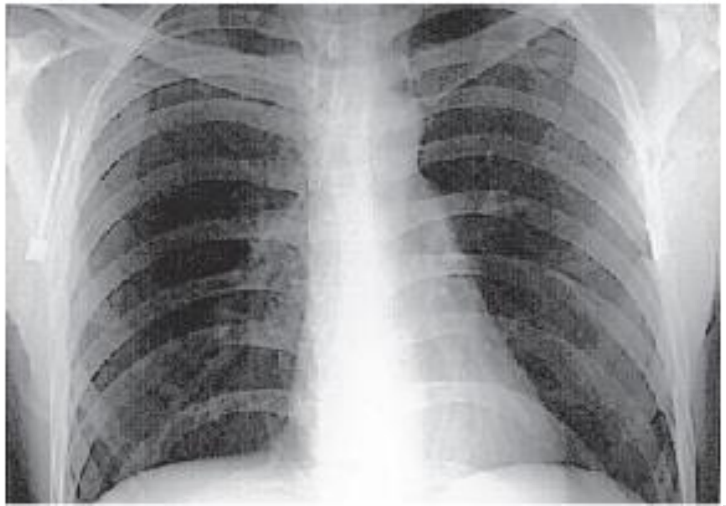
original



GHPF

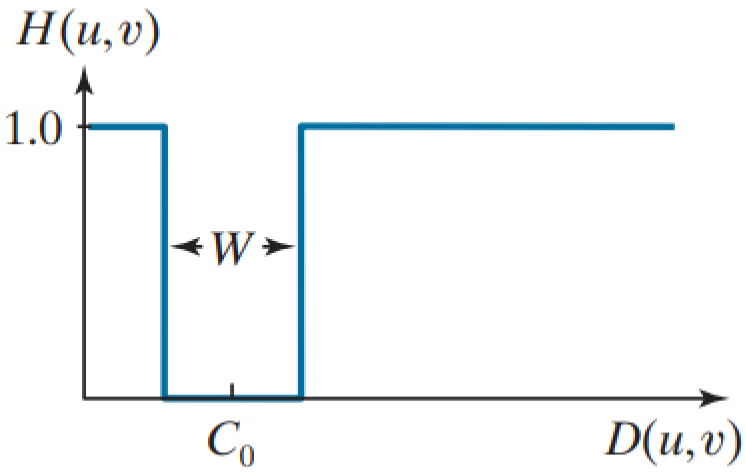
HF emphasis filtering



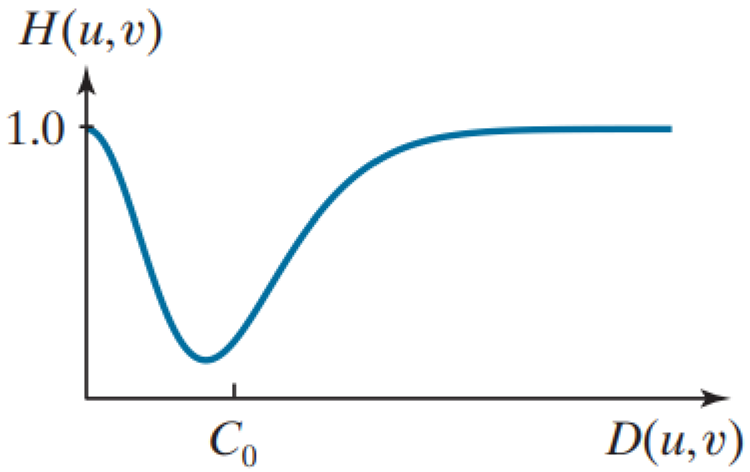


Histogram Equalization

Selective Filtering



Ideal bandreject
A lowpass+a highpass

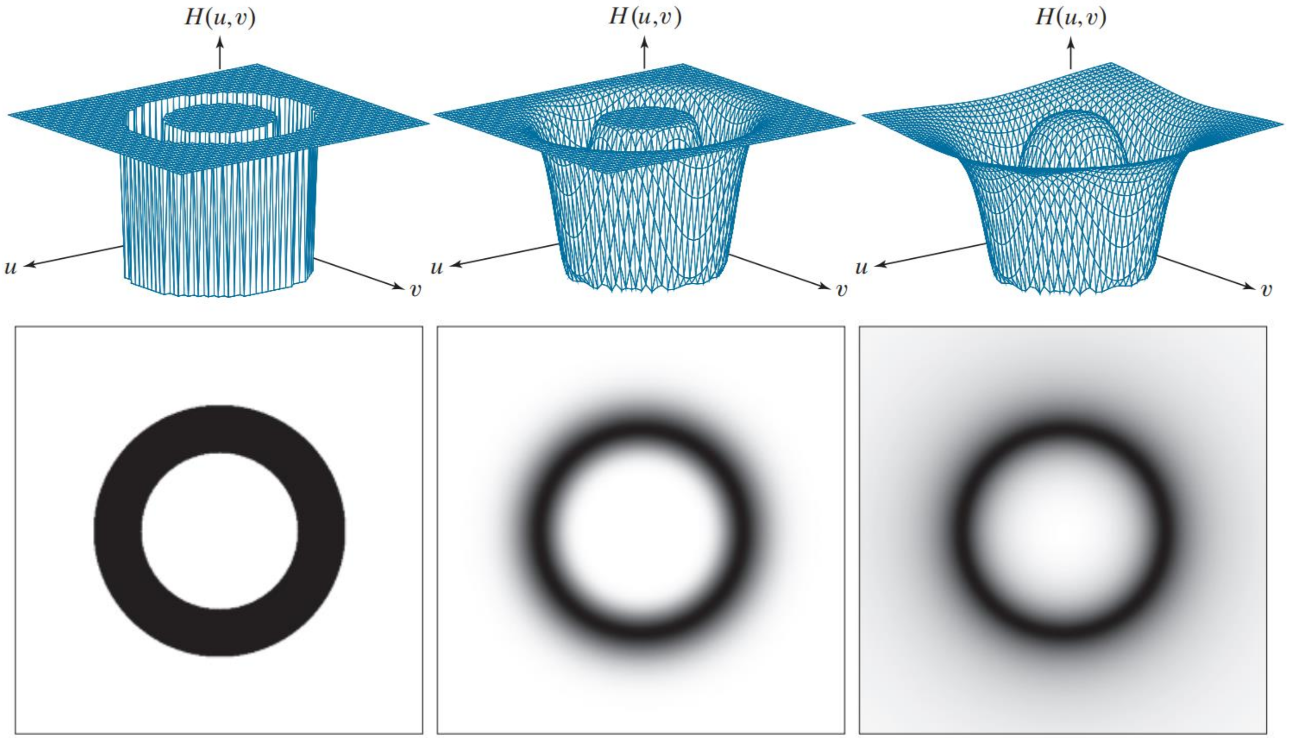


Formed by the sum of Gaussian
Lowpass and highpass filter functions

Selective Filtering

- Ideal, Gaussian and Butterworth

Ideal (IBRF)	Gaussian (GBRF)	Butterworth (BBRF)
$H(u,v) = \begin{cases} 0 & \text{if } C_0 - \frac{W}{2} \le D(u,v) \le C_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u,v) = 1 - e^{-\left[\frac{D^2(u,v) - C_0^2}{D(u,v)W}\right]^2}$	$H(u,v) = \dfrac{1}{1 + \left[\dfrac{D(u,v)W}{D^2(u,v) - C_0^2}\right]^{2n}}$



Selective Filtering

Band-reject and Bandpass Filters

A bandpass filter transfer function is obtained from a band-reject function

$$H_{BP}(u,v) = 1 - H_{BR}(u,v)$$

Notch Filters

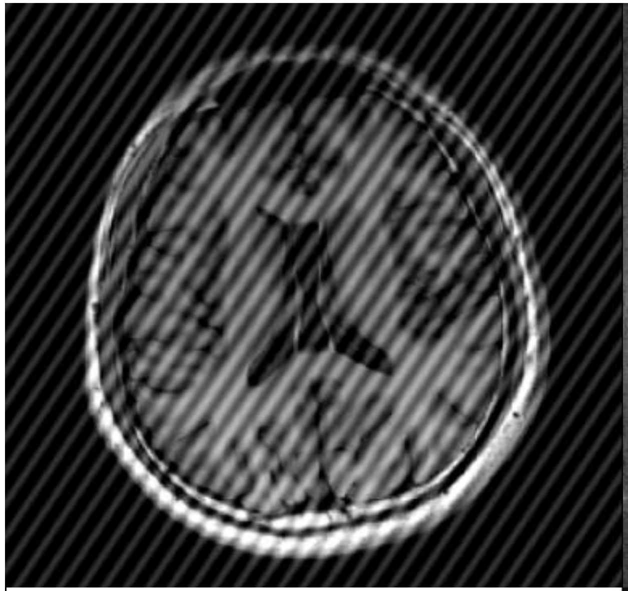
- Notch reject filter are constructed as products of highpass filters whose centers have been to the centers of the notches

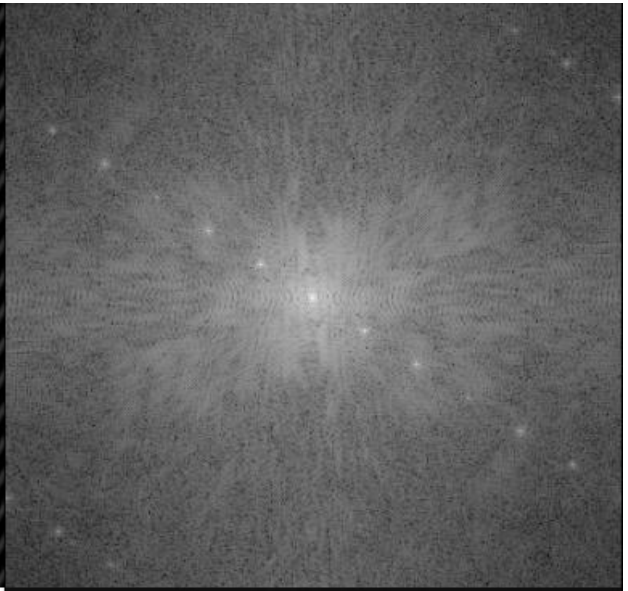
$$H_{NR}(u,v) = \prod_{k=1}^{Q} H_k(u,v)H_{-k}(u,v)$$

Notch pass filter:

$$H_{NP}(u,v) = 1 - H_{NR}(u,v)$$

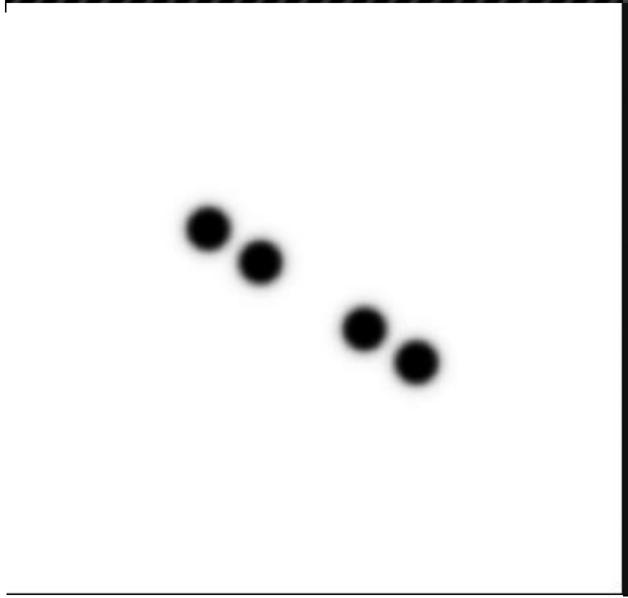
Sampled image showing a moire pattern

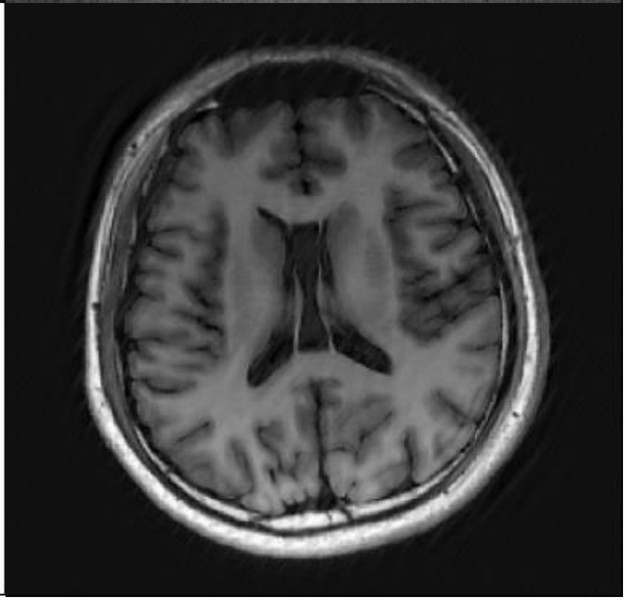




Spectrum

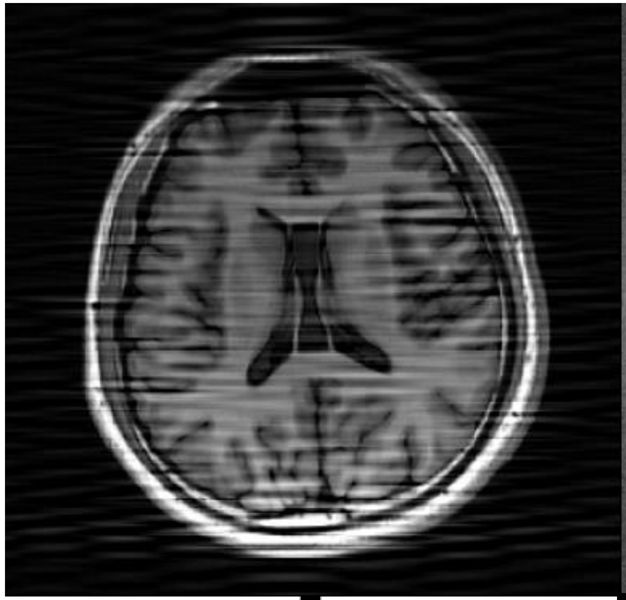
Multiplied by a Butterworth notch reject filter

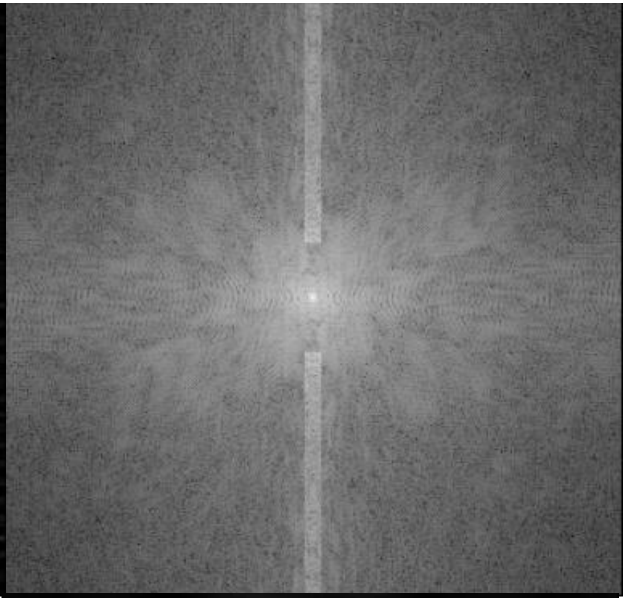




Filtered image

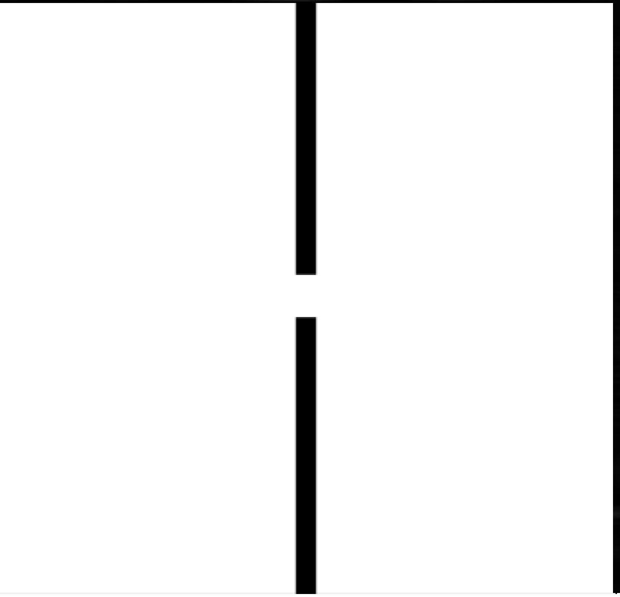
Image of Saturn rings showing nearly periodic interference

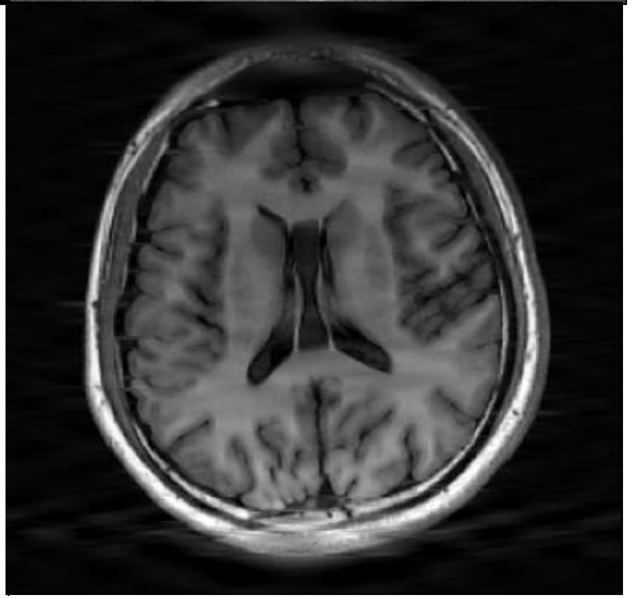




Spectrum

a vertical notch reject filter





Filtered image

Summary

- Frequency Domain Filtering Fundamentals
  - Basic filtering equation
  - Steps for filtering
    - Multiply $(-1)^{x+y}$ (fftshift)
    - padding
- Smoothing: Lowpass Filters
  - Ideal/Butterworth/ Gaussian
- Sharpening: Highpass Filters
  - Ideal/Butterworth/Gaussian/Laplacian
- Selective Filtering
  - Notch Filters